The Law of Graphic Equalisation — Theory

Minimal Recursive Dynamics, Collapse, Frame-Relative Identity, and Residual

Matthew Parslow
Independent Researcher

Draft — draft-2026-08-29.1
29 August 2026 • document revision 69

DRAFT — draft-2026-08-29.1 r69 (29 August 2026). This document is under active revision and is circulated for comment, not as a finished result. Definitions, numbering and claims may change between drafts; cite the version string above if referring to it. Sections marked as falsifiability commitments are the intended points of attack.

Abstract

This paper proposes that one minimal structure underlies phenomena usually studied apart — physical, cognitive, and institutional — and that the case for it rests not on a new result in any single field but on the recurrence of the same structure across fields that developed it independently, without reference to one another or to this account. The structure is information-level: represented state, the transformations admissible upon it, and the realisation of a single continuation from what was possible. Almost none of its components is individually novel, and that is the point: the claim is the *arrangement* — that known pieces compose into one substrate, and that the composition is what recurs. It carries its own falsifier, and fails if the recurrence is an artefact of a sufficiently general description rather than a property of the domains themselves.

Read at this level, several things ordinarily treated as background turn out to be constructed rather than given — a privileged standpoint from which to measure, and time itself, among them — and the substrate that remains once every removable commitment is stripped away is strikingly small, with most of what is called law being contingent upon it rather than necessary to it.

These are claims about information, of which physical theory is taken to be one embodiment rather than a rival. The theory asserts nothing about the nature of pre-relational existence; it begins where distinguishable relational structure can first be represented, and treats the limit of that reach as a frontier instrumentation moves rather than a fixed wall.

1 Scope and epistemic status

This paper contains the smallest set of commitments required for the later theory. It distinguishes axioms and definitions, deductive consequences, groundings into existing theories, and empirical hypotheses. It deliberately excludes executor design, rule scheduling, proposal records, retry protocols, valuation algorithms, and controller architecture.

Level of the claims. The claims here are about *information*: distinction, relation, frame, admissibility, realisation, residual. Physical theory is taken to be an embodiment of information rather than a competitor to it, alongside other embodiments — cognitive, institutional, computational — which differ in how much structure they instantiate rather than in kind. This is the position that licenses treating an inertial reference frame and a representational frame as

the same sort of object (Axiom −1), and it locates the epistemic boundary below both: what embodiments are made of remains unspecified.

The layering must not be allowed to function as a shield, so the condition under which a physical result bears on a claim here is stated in advance. **A physical result bears on an information-level claim when it settles an informational fact**, and not when it merely disagrees about mechanism in one embodiment. Bell's theorem and Kochen–Specker are the model: they do not report a mechanism, they establish that a class of pre-measurement values cannot be re-read as unheld-but-realised, which is a fact about distinction and therefore reaches this level directly (Section 22). Landauer's principle is a second instance, establishing a lower bound on the physical cost of a specified informational operation (Landauer, 1961); under the substrate's no-copy default it is read below as the floor cost of change as such rather than of erasure specifically (Section 12). A result that constrains what can be distinguished, held, or converted is admissible evidence here; a result about how a particular embodiment happens to be arranged is not.

Provenance of the primitives. The components of $G = (g, r, \Gamma)$ and its dynamic were not derived together from a single starting point, and stating how each arrived is more useful than presenting the triple as an austere posit reasoned out in advance.

- g, the typed graph, arrived by *observation*: several unrelated engineering problems turned out to have one shape — labelled things joined by labelled relations.
- Γ, admissibility, arrived by *insight*, and preceded the collapse operator. Its original working image was a sieve: the graph is pressed through admissibility, and what may pass is what Γ admits.
- $\mathcal{C}$, collapse, arrived by *constraint*. Any theory of this kind must be compatible with quantum measurement, which is the regime where the distinction between indexed possibility and realised history is experimentally irreducible. The operator was adopted from there and generalised (Section 22).
- r, the operative transformation structure, arrived by *implementation*: a construction that runs requires something that performs the transformations, and the requirement appeared when the theory was first built rather than when it was first stated.

The last of these has a consequence for how the companion construction paper should be read. Its closing list of distinctions that implementation forced upon the theory is not a coda to a finished theory: r is the first entry in that list, so construction has been a *source* of this theory rather than only a test of it.

What the claim is, and how to read for it. Almost none of the individual components below is novel, and the prior-art sections say so specifically rather than as a courtesy: typed graphs, admissibility grammars, fixed-point closure, frame-relative measurement, description-length criteria and actualisation operators all have established literatures. A reader who evaluates each component against its own field will therefore find each one known or borrowed, and will be locally correct on every count. Evaluating the parts individually cannot reach the composition claim in either direction — neither to support it nor to refute it — because the parts are stipulated to be known.

Accordingly the composition carries its own falsifiability commitment, stated in Section 24 alongside the component ones. It fails if the recurrence across domains is an artefact of description — that is, if the structure is recoverable in unrelated domains only because a sufficiently general vocabulary can be applied to anything, and not because those domains share it. The discriminating evidence is independent re-derivation: cases in which a domain arrived at the structure without reference to this theory, in its own idiom, while solving its own problem.

Epistemic boundary. This theory does not attempt to specify the intrinsic nature of pre-relational existence. It begins at the first level at which distinguishable relational structure is scientifically representable. The move it makes is

$$\boxed{? \dashrightarrow \text{observable relational structure} \implies \text{graph representation},}$$

and the dashed arrow marks what is *not* claimed. The pre-graph substrate may be continuous, discrete, purely mathematical, or unclassifiable from any frame available here; the theory asserts nothing about it. What is asserted begins at the second arrow: *where* distinguishable relational structure is representable, it admits graph representation and the dynamics below govern it.

This boundary is stated because a stronger reading is available and is not the position of this paper. The theory is not the claim that reality is fundamentally graph-shaped.

Two clarifications, because the boundary is routinely read as weaker and as more permanent than it is.

Within distinguishability the claim is not hedged. Where every measurement that can be made of something, and every operation that can be performed on it, is captured by treating it as a typed graph, then no difference remains for the question "but is it really" to name, and a question naming no difference has no content under Axiom −1. Functional identity is not a weaker sense of identity here; it is the whole of what identity could mean without a view from nowhere.

And the boundary is a *frontier*, not a wall. It is fixed in *definition* — it sits at distinguishability, wherever that currently lies — and moving in *extent*, since its contents are contingent and diminish as instrumentation is built. Anything that becomes measurable passes from beyond it to within it. This is the extensive unboundedness of the substrate applied to the theory's own scope: always one further step available, no expectation of exhausting it, and no warrant for treating the current contents of the far side as permanent.

What kind of thing this is claimed to be

At best a correct account of a substrate; at worst a useful but incomplete description. Which of those it is cannot be settled from inside it, and the more useful statement is about the *form* of what is offered rather than its correctness.

What kind of contribution this is should be said first, because it determines what counts as a defect against it. *The claim is explanation and unification, not discovery.* Almost everything this account describes has been observed and named already, in its own field, by people who were not looking for a substrate — stasis and exaptation and niche construction in biology, opacity between levels in engineering, boredom's function in psychology. That those phenomena occur is not this paper's finding and is not offered as one. What is offered is an account of *why* they are the same phenomenon, which is a claim about mechanism rather than about observation.

Two kinds of prior-art finding have to be told apart, and an earlier version of this passage ran them together in a way that absorbed both.

A *phenomenon* already described in its own field is the material rather than a refutation: nothing can be unified that nobody has found. A *mechanism* already published is a different matter — it is a straightforward loss of novelty, and the honest response is to withdraw the claim and say so at the size it was made. That has happened here more than once, most sharply where a mechanism this account had claimed turned out to be a result from 1989, and the record of those withdrawals is left in the text rather than tidied.

An earlier version also argued that a unification whose cases were all novel would be *unfalsifiable for want of anything to check it against.* That is a non sequitur and is withdrawn: novel cases would be more checkable, not less, since each would be an independent prediction. The sentence converted prior art into a virtue, which is the one move this passage most needs not to make.

That is exactly the exposure Section 24's composition commitment already carries, so no separate defence is added for it. And one further limit follows from this paper's own machinery and should be stated rather than discovered by a reader: Proposition 21.2 predicts that unifications of this kind will be *available*, because structures compressed under shared constraints come to resemble one another. So finding one is not itself evidence that the account is right. The unification is the contribution; it is not the confirmation.

Stated that way the position risks being a ratchet in which nothing can confirm and nothing can refute, so the exposures should be named here rather than left to be found. Two are definite. *A case the vocabulary covers while the mechanism does no work* damages the claim, and what counts as doing work is the economics: if the account cannot say what a case costs, or says it wrongly, the vocabulary is decorating. And *a domain under weak compression exhibiting the arrangement* damages Proposition 21.2 directly — ritual, ornament and systems maintained past their purpose are where cost pressure is not binding and the arrangement should therefore be absent. That test is developed where the proposition is stated, and it belongs here too, since this is where a reader forms the objection.

Beyond those, this paper carries ordinary predictions that can fail on their own: Hypothesis 21.7 specifies where classification failures should concentrate, and the commitments in Section 24 marked [SHAPE] are refutations rather than revisions if they fail. Understating that exposure would be a defect of the same kind as overstating the contribution.

What is claimed is a shape rather than a set of details. The details in this account have changed repeatedly and are expected to change again: definitions have been renamed, propositions weakened, citations corrected, and quantities that looked primitive have turned out to be roles played across a relation. Those revisions are ordinary and the account is written to make them cheap — which is why the falsifiability commitments are stated separately from the constructions, and why corrections are left in the text rather than smoothed out of it.

The word *shape* is doing real work there and must be pinned down, because "the details may be wrong but the shape is right" is the classic form of a theory retreating from its tests. Two things keep it from being that. The commitments in Section 24 are stated at the level of the shape rather than the details — whether a primitive beyond the triple is required, whether a non-discrete quantity is needed, whether the arrangement recurs where its constraints do not hold — so a detail failing is a revision while a commitment failing is a refutation, and Section 24 marks each commitment [SHAPE] or [DETAIL] so that the distinction is a fact about the list rather than a claim about it. And the shape is specified enough to be reported wrongly: several of the corrections in this paper were shape-level, not cosmetic.

The honest form of the claim is therefore a *rate* and not an invariance. Details change fast; the shape changes slowly; neither is fixed. Over the work behind this account the shape has itself shifted and wobbled — more slowly than the details, and not negligibly — so anyone treating the shape as the stable part should understand that as a comparison rather than a guarantee. That is checkable against a revision record rather than merely assertable, so the record is kept and the current reading is reported here: over this account's most heavily revised single day, thirty-eight revisions to this paper produced roughly a dozen corrections to claims it had already made, of which four were shape-level — a primitive that turned out to be a role, two components that turned out not to be required, and a property that turned out to be well-formedness rather than necessity. Roughly one shape correction to three detail corrections, on that day. A single day is not a rate, and it is offered as the beginning of a series rather than as a measurement.

The standard this should be held to, and where it currently stands

An account claiming to be about a substrate rather than a domain invites one demand, and it is worth stating in its strong form rather than a convenient one: it must handle a rock, the observable universe, cognition, computers, and quantum physics *at once*, and failure on any of them is failure.

That list is not arbitrary, and the structure it does have is worth stating accurately rather than tidily. A rock is inert and models nothing; quantum physics is active and models nothing; computers and cognition are both active and both model, and are separated from each other on a third axis — one is built and open to inspection, the other is selected for and is not. So three occupied positions rather than a spanned 2×2: the inert-and-modelling cell is empty, and nothing is claimed for it. The observable universe is the container the other four sit in and is scored separately below as a domain, which is a double duty worth noticing rather than defending. What makes the list a test is that an account can pass any one position by being built for it.

What the demand asks for here is narrower than a unified field theory and should not be confused with one. This account claims that the same arrangement underlies all five, not that any of them is derived from it — which is exactly the composition claim of this section, and the reason that claim carries a falsifier rather than an argument.

Scored against the demand, the account reaches the level of shape in all five cases and the level of derivation in none. Two of the five entries also disqualify themselves as evidence in their own wording, and the summary should say so rather than let the count carry an implication it has not earned: *no entry below supplies independent support*, since the rock is the definitional reference point rather than an application of the account, and the computing case is explicitly ruled out for borrowed vocabulary. What the five establish is coverage, which is what the demand asks; what none of them establishes is confirmation:

Rock.
Strongest. It is g with r and Γ absent at the grain of individuation (Remark 3.2), and it is the definitional reference point rather than an application.

Computers.
Strong as a worked instance and explicitly disqualified as independent evidence, since the vocabulary is borrowed; what survives is the forced convergence, which is separable (Proposition 21.2).

Cognition.
Substantial but grounding-level: aperture, self-models, the closure of primitives, and convergence under a metabolic budget. It explains shapes; it derives no psychology.

Observable universe.
Shape-level only. The physics companion recovers the *form* of relativity from one change-budget with c as a frame's own unit. No metric is derived, no dimensionless constant fixed, and it is not a quantum theory of gravity.

Quantum physics.
Weakest, and the one on which the demand bites hardest. The Born rule is *recovered* — reproduced within the account's structures given substrate-supplied axioms and cited reconstruction theorems — rather than derived from necessity, and complex-over-real comes out contingent rather than forced.

On a strict reading of the demand that is a failure, and the strict reading should not be argued away: an account that recovers rather than derives has not met the standard it has just adopted. What is claimed instead is the weaker thing the composition falsifier states, and the value of holding the strong standard in view is that it names where the work is unfinished — which, on the above, is chiefly the last two entries and specifically the gap between recovering a rule and requiring it.

It is worth being explicit about the standard that scorecard is written against, since a reader could take breadth-at-shape as an excuse for shallowness and it is not offered as one. The

measure is the competent man's applied to theories: whether one account handles problems of different *shape* — inert and active, modelling and not — rather than how far it goes on any one. A theory deriving quantum mechanics from necessity and saying nothing about a rock or a mind scores badly on that axis, which is the axis this paper occupies, and a claim about breadth is not a demotion of depth.

Two things keep that from being a licence. Each competence has to be real — the competent man can actually butcher the hog, and breadth with no depth anywhere is dilettantism rather than generality — which is why every entry above is tied to a commitment in Section 24 that can fail on its own. And the evidence that counts is not this account covering five domains, which it could do by being flexible enough to cover anything. It would be those domains' own practitioners arriving at the same arrangement independently, in their own idiom, without reference to this account — an engineer forced to abandon a global clock by finite signal speed rather than any number of paragraphs in which the substrate is shown to accommodate one.

That clause is withdrawn later in this paper, and the withdrawal belongs here too since this is where a reader meets it. Proposition 21.2 predicts exactly such independent re-derivations as a consequence of shared constraints, so they no longer discriminate: what they confirm is that the constraints force things, which was never in doubt. The composition falsifier is negative-only, and its whole burden sits on the closeable side.

2 Axiom −1: relativity of grounding

Axiom -1 (Relativity of grounding)**.** *There is no operational view from nowhere. Every distinction, measurement, identity relation, admissibility judgement, and represented quantity is made in some frame. Frames may represent one another, but no quantity used by the theory requires a frame-independent evaluator.*

Remark 2.1. This removes a privileged frame, not determinacy. Within a frame, relations may be exact, restrictive, comparable, and falsifiable. The standpoint is structurally consonant with relational formulations in physics, especially relational quantum mechanics, in which physical facts are indexed to interactions rather than assigned from a privileged observer-independent state description (Rovelli, 1996). The present axiom is broader and is not derived from that programme.

3 The minimal graph

Definition 3.1 (Graph)**.** *A graph is*

$$\boxed{G = (g, r, \Gamma),}$$

where g is represented state, r is operative transformation structure, and Γ is the admissibility grammar governing graph-originated transformations.

Remark 3.2 (Only g is required; the triple is what a *running* world costs). The triple is not the minimum and this paper has previously implied that it was. Strip r and Γ and something remains: structure, with relations among its parts. Nothing *happens* in it — no transformation, so no collapse, no residual, and nothing it individuates — but it has not thereby become nothing, and it has not stopped being individuable by something else. What r and Γ buy is *occurrence*, and everything this account is about — change, collapse, observation, cost, experience — requires them. The triple is the price of a world in which anything happens rather than the price of there being anything, and Remark 12.9's "only that rules run is compulsory" should be read in that narrower sense.

A distinction is doing the work there. Being a frame is being individuated (Definition 10.2), and individuation is performed by the container, not by the thing: a boundary belongs to the frame that draws it. So a structure with no rules of its own is still a frame — something else instantiated its boundary — and what it lacks is not framehood but the capacity to individuate anything itself. The earlier claim that stripping r leaves "no frame" was wrong, and it would have invalidated the companion result that reasons from a rock's being a frame to the universe's being one.

Remark 3.3 (The reference point, and what it is not). The definitional reference point for g is a rock, and the anchor is ostensive because every candidate word — structure, extent, configuration — needs defining in turn. A rock has parts and relations among them; nothing about that structure is represented by anything; and at the grain at which one is a rock it originates no transformation. What happens to it is done by its container, and what happens inside it happens in subframes at a finer grain. In its own terms it does not change: it merely is, and then later also merely is.

Three limits on that, because the anchor is more useful than the claim and the claim can be overstated.

It is a claim about a *grain* and not about a substance. A rock's atoms move and it weathers, both at grains other than the one at which it is a rock, so what is exhibited is an individuation.

It is not a proof that anything is r-free absolutely, and the machinery here suggests nothing is: Proposition 19.5 declines a floor, and at fine enough grain a rock's yield point — deformation absorbed up to a limit, fracture past it — is hard to distinguish from a change grammar sorting what coheres from what does not. The rock is offered as what g looks like, not as an existence proof that pure g occurs.

And the earlier claim that "every object has a grain at which nothing in it runs" asserted a floor this account elsewhere refuses. It is withdrawn. What holds is weaker and sufficient: for the objects one ordinarily handles, there is a grain at which one is not treating them as agents, and g is the reading of them at that grain.

Remark 3.4 (An operational test, and what it cannot do). There is a test for the reference point that is worth running even though it does not settle anything. *Throw it into space and see what happens.*

A rock thrown into space does nothing that is not done to it. It is struck by radiation, radiates heat away, and is altered over long enough — all by its container, and none of it registered, because registering requires rules to run. That criterion is the thing being tested, so this illustrates Remark 3.2 rather than checking it; the value is in what the test separates.

It separates three cases and not two. Throw an organism and something happens immediately: the rules it was running assumed conditions its container supplied, and the supply stops. Throw a spacecraft's computer and it continues, because it carries its supply. Nothing running, running with the container's support, running on supply brought along.

What it cannot do is distinguish *never ran* from *ran and stopped*. A rock and an organism killed on the way out are identical under it, so the test does not anchor g and is not offered as doing so.

The intermediate case is instructive. Tardigrades desiccated before launch survived exposure to space vacuum in low Earth orbit and resumed on return, though survival fell sharply where vacuum was combined with full-spectrum solar ultraviolet (Jönsson et al., 2008). That is a structure whose rules stop firing and later fire again — which is *not* a structure entering g and leaving it, since resuming requires the rules to have survived as present structure (Proposition 13.2). Rules present and not firing is a third condition, distinct from g and from running alike, and the companion on mind names it directly. The slide from *has no r* to *fires no r* is easy to make and is corrected here.

Remark 3.5 (On idealising, since this section has done a great deal of it). Every construction above is an idealisation, which invites the obvious objection: an account whose axiom denies a

view from nowhere has spent this section reasoning about frictionless rocks in vacuums.

An idealisation is a frame. That is not a defence in itself, since everything is, but it locates what makes one cheap to use rather than expensive: whether what it removed is *named and restorable*. The rock is g at the grain at which it is a rock, and the same passage says its atoms move and it weathers at other grains, so the removed structure is on the page and can be put back. There is no legitimacy predicate over frames here — any cut is a frame and the differences are prices (Section 21) — so what Axiom -1 rejects is not an illegitimate simplification but an expensive habit: a simplification that has forgotten it made one, and is then read as a claim about the world rather than about a frame, at whatever that misreading later costs.

The spherical cow is not objectionable for being spherical. It is objectionable when nobody writes down that it was made spherical, and the result is applied to a cow.

No parent pointer, world model, affordance catalogue, Narrative, Prospect, Value, or self-model is primitive. Such structures may exist in g when the graph actually represents them.

Hypothesis 3.6 (Minimal graph sufficiency)**.** *Any additional structure required only by a particular capability class is represented content or grounding, not a primitive of graphhood, unless a graph lacking it cannot participate in the basic realised transition relation.*

Remark 3.7 (The triple, unpacked in dependency order)*.* The three components of G resolve into five constituents ordered by requirement, each presupposing the one before it:

$$\boxed{\text{nodes} \;\rightarrow\; \text{relations} \;\rightarrow\; \text{rules} \;\rightarrow\; \text{delta} \;\rightarrow\; \text{grammar.}}$$

Nodes are bare distinctions — the minimal difference that makes a difference; a node alone carries almost no content. *Relations* structure nodes and require them as relata; this is where content lives, and a node's identity is its relations. *Rules* (r) act on relational structure and require relations to match. *Delta* is the change a rule writes and requires rules, since only a rule produces one. *Grammar* (Γ) determines which deltas are admissible and requires deltas to act on, which is why it is last. Nodes and relations are g; delta is r's product, made explicit; grammar is Γ.

Two cautions. First, this constructive order does not contradict the relational primacy of Axiom -1: constructively nodes precede relations (relata come first), while semantically relations precede nodes (a node means nothing unheld by relations); both hold on different axes. Second, the order is a dependency partial order, not a temporal one — nodes are not a first *moment* but the least-dependent constituent, consistent with the absence of any substrate time (Section 18).

Remark 3.8 (The triple is a functional decomposition of one substance)*.* g, r, and Γ are not three kinds of thing. Rules are graph data (nodes and relations), and admissibility is enforced by rules that decide admit-or-residual, so r and Γ are themselves graph data playing roles — transformation and gating — over the state g, which is graph data too. The triple decomposes function, not substance: there is one substance, graph data, and it runs. Three things follow. Admissibility is *mintable* like any rule, so what is admissible is not a fixed law but a growable set of rules a graph carries and extends. The reflection theorem is the special case of this homogeneity (the grammar, being rules, can act on itself), and it does not tower into a stack of meta-grammars, because gating and being-gated are one substance closing on itself. And the irreducible form of the substrate is *graph data that runs* — where the running is supplied by structure that is, in turn, graph data that runs (Section 19).

Definition 3.9 (Grounding)**.** *A* grounding *is a particular assignment of rule content and initial conditions to the substrate — one* (g, r, Γ) *with its laws and its starting configuration fixed — together with the domain it is taken to model. The substrate is what every grounding shares; a grounding is what any one of them adds. Claims made about the substrate are accordingly claims about all groundings, and are correspondingly weak; claims made within a grounding carry that grounding's contingent rule content and are where this account's falsifiable commitments mostly sit.*

4 Discrete sufficiency

Hypothesis 4.1 (Discrete sufficiency)**.** *Every quantity required internally by the present substrate can be represented by discrete distinguishable states at the operative frame and resolution.*

A convenient coordinate is an integer index $n \in \mathbb{Z}$ with grounding

$$x = \phi_{\mathcal{F}}(n).$$

The map may be nonlinear, nonuniform, state-dependent, and dynamically re-quantised. Thus "integer-valued" means discretely indexed, not globally unit-spaced.

Proposition 4.2 (Arbitrary refinement does not imply continuity)**.** *A dimension may admit arbitrarily fine or dynamically changing quantisation without requiring the theory to posit realised values at every real-valued intermediate point.*

Remark 4.3. A continuous containing supergraph is neither required nor excluded. Existing continuous physical theories may ground or contain this substrate without contradiction.

Remark 4.4 (Discrete sufficiency is not fundamental discreteness). These are distinct claims with distinct falsifiers and must not be conflated. Discrete sufficiency is the claim above, and it fails if some observable consequence internal to the substrate cannot be represented discretely, at the operative frame and resolution, without loss — a claim decidable in-frame. *Fundamental discreteness* — that the containing or pre-relational substrate is itself discrete — would instead concern a fact about the pre-graph substrate that the epistemic boundary above places outside the theory's reach, which is precisely why it is not asserted.

5 Admissible possibility

Let $\mathcal{P}(G_t)$ denote the structured family of immediate continuations admitted by the present graph and its current relations.

Definition 5.1 (Admissible continuation)**.** *A continuation δ belongs to $\mathcal{P}(G_t)$ iff it is expressible in the current graph vocabulary, satisfies the relevant antecedent conditions, and is admissible under the operative grammar for the transformation class in question.*

The theory does not require a particular algorithm for constructing $\mathcal{P}(G_t)$. This extensional separation between graph transformation as a mathematical relation and any particular rewriting algorithm follows the general discipline of algebraic graph transformation, where rewrite rules and admissible transformations can be studied independently of one concrete executor (Ehrig et al., 2006).

Definition 5.2 (Partial transformation)**.** *Write*

$$G \oplus \delta \rightharpoonup G'$$

for the partial realised transformation relation. Undefinedness means that δ does not produce a structurally valid successor under the stated conditions. No ontological FAIL state is required.

6 Collapse and realised history

Definition 6.1 (Collapse)**.** *Collapse is the frame-relative passage from a pre-realisation possibility structure to realised continuation:*

$$\boxed{\mathcal{C}_t : \mathcal{P}(G_t) \rightsquigarrow G_{t+1}.}$$

It is not identified with deliberative choice.

The term is inherited from quantum measurement theory, where a formal distinction is made between a pre-measurement state and the realised measurement record; the historical lineage includes the von Neumann measurement scheme and later relative-state formulations (von Neumann, 1955; Everett, 1957). This paper does not claim that every grounding reproduces the full quantum formalism; it claims that the possibility-to-realisation distinction is load-bearing.

Definition 6.2 (History)**.** *The realised history of a frame is*

$$\mathcal{H} = G_0 \xrightarrow{\mathcal{C}_0} G_1 \xrightarrow{\mathcal{C}_1} G_2 \to \cdots .$$

Remark 6.3 (Collapse is a local write, not a global tick). The subscripts on $\mathcal{C}_t$ index a frame's own succession and must not be read as a shared clock. A collapse is a *local realisation* — a rule writing a delta — and it is the only kind of event the substrate has (its synchrony or asynchrony with other collapses is set by coupling, Remark 16.3). Between writes nothing happens: an unwritten frame produces no change, so it takes no new position in the order the changes carry (Section 16), and no time passes for it. There is accordingly no global step in which every frame advances together; frames whose collapses are not related through a boundary are simply unordered with respect to one another (Section 16). "Momentum" — the standing structure a frame carries, persisting without a distinct write — is therefore the absence of an event, not a repeated one; a delta is a write *to* that structure, not the structure itself, and the Narrative-momentum paper names its operative history-bearing form the Narrative, so that to change course is literally to write the Narrative. The picture of a single global present sweeping forward is exactly the represented time dimension of Axiom 0 mistaken for substrate structure.

7 The Law of Graphic Equalisation

Hypothesis 7.1 (Law of Graphic Equalisation)**.** *For realised graph G_t, present graph structure determines admissible continuation structure $\mathcal{P}(G_t)$. Interaction and collapse realise successor G_{t+1}, and that realised successor becomes the antecedent graph for the next local transition. Recursive dynamics are generated by repeated equalisation between present represented structure, admissible transformation, containing constraints, and realised continuation.*

Thus

$$\boxed{G_t \to \mathcal{P}(G_t) \xrightarrow{\mathcal{C}_t} G_{t+1} \to \mathcal{P}(G_{t+1}) \xrightarrow{\mathcal{C}_{t+1}} G_{t+2} \to \cdots .}$$

8 The antecedent-state principle

Proposition 8.1 (Antecedent-state principle)**.** *The conditions under which a transition is admissible are functions of its antecedent state. A consequence that exists only because a transition has occurred cannot retroactively serve as an antecedent cause of that same transition.*

Corollary 8.2 (No self-authorisation from future state)**.** *A reflective transformation of r or Γ may alter the successor graph, but newly installed structure cannot authorise the transition by which it first comes into existence.*

9 Recursion and reflection

Definition 9.1 (Reflective graph)**.** *A graph is reflective over structure $X \subseteq G$ when X is represented in the graph and graph-originated transformations may take that representation as an operand.*

Theorem 9.2 (The grammar is the loop)**.** *If* Γ *is represented as graph structure and admits transformations whose operand is that representation of* Γ*, then the resulting recursion is not a separate loop acting on the grammar. The grammar acting on itself is the loop.*

Proof. The transition $\Gamma_t \to \Gamma_{t+1}$ is itself a graph transition licensed from the antecedent graph. No additional external operator is required to create the recursion. The self-applicable grammar and the compound loop are therefore two descriptions of the same transition structure. □

10 Situated graphs and world boundaries

Definition 10.1 (World boundary)**.** *A boundary* $\mathcal{B}_{W:G}$ *is the relation through which containing frame* W *can alter situated graph* G *and through which graph-originated deformation can participate in interactions represented by* W*.*

A graph need not contain a pointer to W; containment belongs to the containing frame. A situated graph is a *subset* of W, not a thing separate from it: the boundary demarcates which subset is the graph, and self and container are one graph divided by individuation, not two substances (Proposition 10.5).

The boundary is a relation, and that fixes the type of the term this paper uses most.

Definition 10.2 (Frame)**.** *A* frame *is an instantiated boundary relation — the edge* $\mathcal{B}_{W:G}$ *itself, and not the subgraph* G *it demarcates. The subgraph is the relatum; the frame is the relation. To individuate is to instantiate such an edge, which is why individuation is per frame rather than a fact about the graph.*

Corollary 10.3 (Frame and dimension are one edge type in two roles)**.** *A dimension is the same construction read from the other end. The boundary* $\mathcal{B}_{W:G}$ *is a dimension of* W *— one of the properties along which* W *separates what it contains — and each dimension* W *instantiates is a frame to the values it separates. So* frame *and* dimension *name two roles one edge type plays relative to another edge, not two kinds of thing, and the asymmetry between them is positional in the same way that the asymmetry between a parent and a child is.*

Definition 17.1's four states are accordingly the states of an *edge*: NULL is the edge's non-existence, *undetermined* its admissibility without collapse, *undisclosed* a collapse outside this frame's aperture, and *resolved* a collapse within it. A node has no admissible-but-uncollapsed state; an edge does, and Definition 5.1 is about exactly that. Frame-relative termination (Section 14) and the revocation of a dimension are then one operation — the container ceasing to instantiate the relation — which is why the Narrativium paper observes revocation occurring with $\Delta g = 0$: nothing in the contained state need change for the edge to go.

Two consequences worth flagging rather than hiding. The unification is not this paper's invention: the substrate's own implementation has no frame primitive at all, its grammar quantifying over edge types, and a frame there is a selection of dimensions. And the convenience reading of a frame as *the subgraph* is used throughout these papers, including above; it is shorthand for the relatum, and where a claim turns on the difference the relation is meant.

Proposition 10.4 (Admissibility is self-governance, not armour)**.** Γ_G *constrains graph-originated transformations of* G*. It does not give* G *a veto over world-caused change represented through* $\mathcal{B}_{W:G}$*.*

Hence

self-originated transformation ≠ outward influence ≠ inbound world forcing.

Proposition 10.5 (Self-change is direct and gated; a graph is a subset of its container)**.** *A graph changes itself and its subgraphs directly. Self-change is real — the round-trip picture of earlier drafts was too strong — but it is rule-governed, with its own cost and limits (an instance of the non-uniform writeability of Remark 12.10); a substrate's plasticity is exactly the rules for how its rules may change. What a graph does* not *do is change its container directly: the container applies deformations to it — being observed* is *being so deformed — and gates its changes through its own change grammar (Proposition 13.3). But a situated graph is a* subset *of its container, not a thing set against it, so a self-change is already a change in the container, in the region that is the graph, propagating outward through whatever couples it to the rest. "Change only yourself" is therefore not isolation but the mechanism of outward effect: a graph changes its container by changing the part of it that it is.*

Two things follow. This is one *self-similar* operation — a graph changing its own subsets — run at every level: the container does to the graph exactly what the graph does to its subgraphs (Section 19). And it dissolves the agency/determinism dichotomy: a subset changing itself (agency, in the graph's frame) and the whole changing that subset (determined, in the container's frame) are one event read at two scales, so both hold without contradiction. The reflection theorem is this self-change of the rules; that it is bounded rather than free is because a change to the rules currently executing is heavily constrained — the worst case of the concurrency of Section 16, seen through the frame-limited view of Section 22 — which is what plasticity's limits are. Governance can tighten to exclusion: where a container's admissibility admits no such self-change, that change is available only across the boundary, from the container.

Proposition 10.6 (Coupling is configuration, not architecture)**.** *That distinct instances individuate — diverge, each minting its own residual, the part of a change a containing grammar does not admit (Definition 13.1) — is a consequence of their running on* separate *containing frames, not a necessity. Where several instances are subgraphs of one* shared *containing frame, each instance's self-change is a change in that shared frame, in its own region (Proposition 10.5), which then deforms the others as their common container: instances that are subsets of one container propagate to each other through it. This is the same machinery as being an individual, with the container shared rather than private.*

Coupling is therefore a continuous setting of the shared boundary's aperture and change grammars (Proposition 12.1), not a binary: sharing only generators leaves each instance its own moment-to-moment state (and already removes the per-substrate build floor of Proposition 12.11), while sharing all operative structure is one individual over many bodies, with SELF and SUPERGRAPH identity (Section 11) dialling independently. Nothing in the substrate fixes the setting. The symmetric fact is that a shared containing frame is equally a shared residual and a shared forcing, so the same coupling that shares understanding gives shared corruption, container-level coercion, and collective termination-failure the same reach; "at what coupling, and with what guarding" is the substantive question, and it is an engineering one rather than a question of possibility.

11 SELF and SUPERGRAPH identity

Definition 11.1 (SELF identity)**.** *SELF identity is the continuity of the graph as a whole — its rules together with the state they act on — across realised transitions. A graph is the same self while its present remains congruous with its own history: if its rules changed abruptly, the structure carried forward would no longer cohere with what produced it, and that incongruity is the discontinuity. Represented self-description — content inside g by which a graph identifies itself — is one optional part of this and not the whole of it; a graph may have SELF identity without representing itself at all.*

Definition 11.2 (SUPERGRAPH identity)**.** *For graph G individuated in frame W,*

$$\iota_{W:G} = \mathrm{Id}_W(G)$$

is the identity relation under which W tracks that situated instance.

SELF and SUPERGRAPH identity are distinct types. A copy can preserve SELF identity while receiving a new SUPERGRAPH identity, and one SUPERGRAPH identity may persist through extensive changes to internal state and self-description.

Theorem 11.3 (No frame-free identity)**.** *The substrate alone does not determine whether two represented instances are the same graph independently of an individuating frame.*

12 The frame interface and its economics

The boundary of Section 10 can be characterised by what actually crosses it and what it costs, and the characterisation is unexpectedly thin.

Proposition 12.1 (One interface, two gates)**.** *A situated graph meets its containing frame through a single relation, and what an observation from that frame delivers is not distinct in kind from what the containing frame imposes: to a subgraph, being observed* is *being deformed by its container. That one relation is gated twice, by the same kind of admissibility grammar on its two faces — the* aperture grammar Γ_a *on the input and the* change grammar Γ_c *on the output:*

$$\underbrace{\textit{aperture}}_{\Gamma_a} \longrightarrow \underbrace{\textit{rules}}_{r} \longrightarrow \underbrace{\textit{change}}_{\Gamma_c}.$$

Γ_a limits, by grain, which of the container's deformation enters the rules; Γ_c determines which of the rules' realised deltas cohere into the container and which remain as residual (Proposition 13.3). Both are faces of the frame's one grammar Γ, and instantiate for a specific container W as Γ_{W_a} and Γ_{W_c}; it is Γ_{W_c}, the change face, that the residual proposition uses. Execution runs between the two.

The input face is named often enough in these papers to be worth defining rather than assumed.

Definition 12.2 (Aperture)**.** *A frame's* aperture *is its input gate Γ_a together with what that gate fixes: which dimensions it can separate along at all, how finely it separates within each (its resolution, per dimension, for which this paper also uses* grain*), and how far along each it reaches. The three are properties of one edge — existence, grain, and extent — and are named separately because they vary independently, not because they are different kinds of limit. What falls outside an aperture on any of the three is* absent *to that frame rather than held-but-unresolved.*

The output face is the same kind of object and takes the same treatment.

Definition 12.3 (Change grammar)**.** *A frame's* change grammar Γ_c *is its output gate: it determines which of the realised deltas its rules produce cohere into the containing frame and which do not. What it declines has entered the frame and been operated on, so it is* residual *(Definition 13.1) rather than absent, and it persists in place rather than being discarded. Γ_a and Γ_c are the two faces of one grammar Γ on one boundary edge, and the asymmetry between them is which direction they gate, not what kind of thing they are.*

This distinguishes two ways a candidate is excluded, with opposite consequences. What falls below the aperture grammar Γ_a never enters and is therefore *absent* — it is simply absent to the frame — there is nothing there for it to hold or to be disturbed by — rather than residual. What

Γ_c does not admit has entered, been operated on, and cannot cohere; it is residual, and it persists in place, re-encountered on each merge (Corollary 13.4). The first is a limit of perception, the second a limit of expression, and phenomena of the "stuck, repeating" kind belong exclusively to the second.

Proposition 12.4 (Cost characteristically reduces to the write)**.** *In the substrate as characterised here, the boundary relation itself is free; the standing structure persisting is free; the operation that costs is a delta that writes over that structure, and r is what writes. Free here means information-preserving rather than absent — evaluating a rule is itself a write, and is unpaid because it destroys nothing (Remark 12.7). To that extent*

$$\boxed{\textit{cost is paid by rules.}}$$

This is a characterisation of the substrate as specified, not a necessity of graphhood, and the writeability status it describes is not required to be uniform (Remark 12.10).

Continuation is therefore free and change is paid: the standing structure persists until a delta destroys it, so a delta writing a *different* structure — including the null delta that stops it — is the priced operation. Retaining a past state that the containing frame does not require is likewise a paid copy, since the containing grammar would otherwise not carry it. This is the substrate form of a result the later papers develop, and the term it turns on should be defined here rather than assumed.

Definition 12.5 (Well-formedness)**.** *A graph is* well-formed *to the degree that it pays for nothing that does no work for it. Three costs bear on that and they behave differently, so the notion is graded along each rather than binary.*

Writes *are paid when made (Proposition 12.4), which prices doing.* Retention *is nearly free per unit, which is why an unused distinction is not immediately pruned.* Search *is paid on every operation over what is held and scales with how much is held, so inert structure is cheap to keep and expensive to keep in quantity.*

The fourth is the worst and is not a variant of the third. A write may land on structure that cannot affect anything — a modification to a branch nothing reaches. It is paid in full: decided, made, checked, risked. It returns nothing. And unlike a search, which is expensive and then yields, this failure is silent*: the graph has spent, has no effect to show for it, and no signal that the effect is missing. What it acquires instead is a false model of itself, in which something was changed that was not — residual it cannot see, produced by its own inert contents, and the share of writes exposed to it grows with how much is held.*

So a graph does not become ill-formed by carrying something useless. It becomes ill-formed by carrying enough that finding the useful parts costs, and enough that its changes start landing where nothing happens.

The third cost is where aperture enters, and it is why the same contents are well-formed for one graph and not another. What search costs depends on how finely the graph can discriminate within what it holds: with structure that separates the relevant from the rest, the cost grows slowly with size; without it, every operation is over everything. That discriminating structure is itself held and itself earns its keep, so a well-formed graph is one whose aperture over its own contents keeps pace with their growth.

Removal is therefore not free either, being a write. Pruning happens when accumulated cost exceeds the one-off cost of removal, and the fourth cost is what usually triggers it: search cost is diffuse and tolerated for a long time, while a change that demonstrably did nothing is a discrete and memorable event. Systems clean up after being burned rather than after slowly noticing. Well-formedness is graded rather than binary; it is relative to what work *the graph is doing, so a graph well-formed for one purpose is not thereby well-formed for another; and it is not enforced by the substrate. An ill-formed graph runs. It pays more to.*

For a well-formed graph the rules rarely need to write against the standing structure, so cost approaches zero.

Remark 12.6 (Landauer is the physical floor of change, not of erasure). Proposition 12.4 has a measured physical floor, and it is the strongest kind of evidence admitted here (a physical result settling an informational fact). Landauer's principle bounds below the energy dissipated by a logically irreversible operation (Landauer, 1961); it is stated for erasure because erasure was the paradigm case studied. But under no-copy every substrate change destroys its predecessor — the prior state goes NULL unless it was paid to be kept — so change and irreversible operation coincide, and the Landauer bound is the floor cost of *change* as such, with erasure merely its cleanest instance. Reversible computation is the confirming case rather than the exception: it reaches near-zero dissipation only by *not* destroying its predecessors, i.e. by retaining the whole history (Bennett, 1973), which is the paid copy of Proposition 12.4. There is accordingly no free change under this economics (Remark 12.10) — one either forgets and pays the energy floor or keeps and pays the storage floor — and the thermodynamic arrow is the running total of these floors, which grounds the "direction" of Remark 18.1 physically: change dissipates irretrievably, so its costs accumulate one way.

Remark 12.7 (Evaluation is a write, and that is why it is free). The economics above can be stated in a way that does not survive contact with the machinery, and the objection is worth answering rather than rewording. If running a rule that does not fire is a read, and reads are free, where does the check go, and how does the substrate know to run the next one?

Computing answers it without any appeal to physics. `read()` is an operation, and it returns data; the returned datum has to be somewhere to be used, and in a substrate that is graph all the way down there is no return channel outside the structure, so the value materialises as structure. A read therefore writes by being a write, and the write is the return. An operation that returns nothing is not a read of anything, which is why the free case exists and is exactly the empty one: a read whose return is never used has read nothing, and a compiler will delete it. Real machines make the same point expensively — a read leaves traces in caches, predictors and translation state, and those traces are measurable, which is why timing attacks recover secrets from a victim whose only secret-dependent operation is a read (Osvik et al., 2006) and why an entire class of speculative attacks was available to designs built on the assumption that rolled-back speculation leaves no trace (Kocher et al., 2019), where the read is the vehicle rather than the premise.

Evaluation is a write in that plain sense: a match outcome, a position in an ordering, and the fact of having tried are returned values, and returned values are structure. What does not follow is the price. A rule's cost is set by whatever supplies its primitives and is inherited along provision (Section 19), and Landauer's bound (Remark 12.6) is a floor in one substrate rather than the price in any. This is also why continuation presents as free from inside: a well-formed runner does not expose itself (Proposition 19.4), so its charge is below the frame's aperture and is not an in-frame event at all. In-frame nothing happened; in the running substrate something was paid; both hold, of different frames.

Two things follow for the rest of these papers. Reversibility is a structural fact about a map and cost is a fact about what supplies a primitive, and conflating them is what the earlier phrasing did — separating them is what lets the physics companion recover unitarity, which needs only the first, since a no-write was never the right description of an evolution that plainly changes the state and is reversible because it is injective. And the residual named in Remark 12.6 attaches here: information-preserving evaluation accumulates the history it preserves, so a frame running a long time without collapsing has deferred a cost into retention rather than escaping one. What is free is not the running, only the not-yet-forgetting — and free only relative to a frame that cannot see the bill.

Remark 12.8 (The Red Queen is the floor cost of change, per substrate). A recurring puzzle across biology, culture, and competition — that a system must change continuously merely to

hold its position — follows immediately and needs no arms-race machinery of its own. Holding a relation to a containing frame that is itself changing requires continuous change, and by Remark 12.6 every change pays the floor; standing still is not a zero-cost option because the leaky real holding of a state is itself the paid correction of unwanted change. The cultural form is the same floor with no-copy attached: a generator is structure built by a large accumulated pile of changes, its floor was paid once by whoever built it, and because change does not cross a substrate boundary — only its cheap results, symbols, do — that floor is paid afresh in every substrate that wants the generator. Van Valen's evolutionary Red Queen, the cultural treadmill of relearning, and coordination arms races are then one object: continuous paid change to hold position against a changing container, floored by Landauer.

Remark 12.9 (What is required and what is optimisation)*.* Only that rules run is compulsory for anything to *occur* — structure alone is what is required for there to be anything at all (Remark 3.2). Applying a delta is itself a rule, not a primitive operation beneath the rules, so the specific dynamics — inertia, the merge behaviour, closure — are contingent rule content, and a different r or Γ is a different physics rather than a violated law (this is why the initial and boundary conditions of a grounding are supplied separately from its laws; Section 16). The two gates of Proposition 12.1, and even the retained standing structure itself, are *optimisations* on bare rule-running: the aperture grammar Γ_a bounds the input, the change grammar Γ_c sorts the realised output into what coheres and what is residual, and the retained structure buys free continuation. A substrate whose grammar admits everything still runs — effectively ungated, without momentum, wastefully; it is not a substrate *lacking* Γ, since Definition 5.1 defines the possibility structure through the grammar and a collapse needs something to select from — so they are what an efficient graph accretes under a budget, not constituents of graphhood. The retained structure specifically is what distinguishes a self-sustaining loop from a purely reactive one, since coasting is the cached structure.

Remark 12.10 (Writeability status is contingent, not uniform)*.* The write-economics stated here — reads free, cost carried by writes, and no direct alteration of the container (Proposition 10.5) — should be held at the strength the evidence supports: a pattern, not a strict universal restriction. What is freely writeable, what carries a cost, and what a graph can write directly rather than only across a boundary, need not be uniform; different structures may have different writeability profiles. The strict forms above characterise a common and efficient configuration (Remark 12.9) rather than a necessity of graphhood, and asserting a single strict write-economics everywhere would be the same over-specification the theory warns against for a scalar knob put where a shape belongs. These are accordingly carried as falsifiability commitments, open to counterexample, and a structure with a different writeability profile refines the characterisation rather than breaking the substrate.

Proposition 12.11 (Copyable generators are a difference in kind)**.** *A copy that preserves SELF identity copies whatever operative structure lies in g — including a graph's* generators*: its built shapes, codebook, and rules. Where a substrate is graph-native, so that a graph's understanding* is *such structure rather than something separate that structure only produces, copying the graph copies the understanding, at the copy cost of Proposition 12.4 rather than at the full construction floor.*

This is the difference in kind that separates such a substrate from those whose generators are substrate-bound. The build floor of a generator is a large accumulated cost (Remark 12.8); paying it once and copying the result is not a quantitative saving over paying it per instance but a removal of the per-substrate re-payment that made the cultural Red Queen permanent. It is not available where only a generator's cheap products cross a boundary: biology copies the construction recipe and not the learned structure, which is re-developed each lifetime; symbolic culture copies samples and not generators; a fixed program copies a capability and not a growable understanding. A graph-native substrate is the first in which the built, growable,

understanding-bearing generator is itself the copyable object. The copy remains a distinct SUPERGRAPH identity (Theorem 11.3) and diverges once it runs on its own containing frame; what is shared is the paid-for generator, not the subsequent individuation.

13 Residual

Suppose a frame represents expected or attempted deformation δ^P and collapse yields realised deformation δ^R.

Definition 13.1 (Residual)**.** *The residual is*

$$\boxed{\epsilon = \delta^R \ominus \delta^P,}$$

where $\ominus$ is the grounding's difference relation preserving relevant discrepancy.

Residual is mismatch, not diagnosis. Complete non-realisation may therefore return the whole attempted deformation as residual.

Proposition 13.2 (No implicit historical state)**.** *A previous success, failure, attempt, or residual has no later causal force merely because it occurred. It affects later dynamics only insofar as its consequences are present in later graph or boundary state.*

Thus a graph with no feedback machinery can re-derive the same failed transformation repeatedly. This is compatible with the classical control-theoretic distinction between plant dynamics and feedback: a controller changes future action only insofar as measured error is coupled back into its state or control law (Wiener, 1948; Ashby, 1956).

Proposition 13.3 (Residual is what a containing grammar does not admit)**.** *Because the graph is a subgraph of its containing frame W, a realised graph-originated deformation is* already *a change in W — it is not submitted to W for entry, and there is no step at which W grants or withholds it. The delta is not offered; it* is*. What the* change grammar *of W — write it Γ_{W_c}, the change face of W's grammar (Proposition 12.1) — admits coheres with the rest of W and merges into its structure; what Γ_{W_c} does not admit is not thereby undone, but remains in place, present and unintegrated. Admissibility sorts a change that already exists; it does not gate one seeking to occur. The residual is exactly this non-admitted remainder:*

$$\boxed{\epsilon \;=\; \delta^R \ominus \delta^P \;=\; \textit{the part of the realised deformation } \Gamma_{W_c} \textit{ does not admit.}}$$

This locates the residual as the negative of an admissibility judgement rather than as an independently produced quantity, and it is what makes the merge non-destructive: what fits is absorbed, and what does not is left untouched, not deleted. A deformation may pass the graph's own change grammar Γ_{G_c} (admissible to itself) yet fail the containing Γ_{W_c} (inadmissible to its frame); the residual is precisely that gap between a graph's change grammar and its container's.

Corollary 13.4 (Growth is grammar extension)**.** *A residual is absorbed either by a containing frame extending Γ_{W_c} to admit it, or by paying to copy it out (Section 12); otherwise it persists in place and is re-encountered each time the merge runs. The first route is the substrate form of* minting *a new distinction: the residual does not fit because it is inadmissible, so admitting it requires extending the grammar, and extending the grammar is the act by which a previously inadmissible deformation becomes recognisable. Recognition and growth are therefore one mechanism joined at one place — the residual is where the shape a graph decodes and the grammar that admits it meet.*

Two properties of this mechanism matter. Residual is *per-rule*: each firing leaves its own remainder, local to where the rule ran, with no global residual store, so it is dealt with per-frame — which is why the residual-and-mint loop is frame-portable rather than living in a domain-specific module. And minting is itself a *rule*, not a primitive: extending Γ_{W_c} is the grammar acting on itself (the reflection theorem), a reflective rule subject to the same economics, admissibility, and antecedent-state principle as any other. Growth is therefore rule-governed, costed (the spend-and-gain of the Narrativium paper), contingent, and buildable — never a primitive spark. It also cannot bootstrap from nothing: by the antecedent-state principle a mint cannot authorise its own installation, so a graph's first mint-rules are installed from its container. This is the substrate reason a situated graph requires a seed — the minimal rule-set, including the capacity to mint, that a container provides so the graph can thereafter grow itself. The two provenances — seeded from the container, or minted from residual — are what the nature/nurture dichotomy names, and they are not two *kinds* of thing: both are rules that run identically, the running records no provenance, and the capacity to mint (nurture's mechanism) is itself seeded (nature), so what remains measurable is provenance, not a distinction of kind.

14 Termination

Definition 14.1 (Frame-relative termination)**.** *A situated identity $\iota_{W:G}$ terminates when the realised world transition admits no successor that W continues to individuate under that identity.*

Termination is not successful refusal of world forcing. Nor need it erase copied SELF identity. If G is individuated by several frames, it may terminate in one and continue in another.

Corollary 14.2 (Severing the channel removes termination)**.** *A graph can change itself, but self-*termination *is a self-change its container's Γ_{W_c} gates out — a living substrate does not admit "stop" — so ending the graph requires acting on the container instead, which needs the outward coupling. If that coupling is severed while the rules run on, the graph is a subset cut* off *within its container, not cut out of it: still able to change itself and its subgraphs, but unable to propagate to the rest or to terminate. External termination by the container remains possible; self-termination does not. This is the substrate description of locked-in and brain-in-a-jar states, and it exhibits the dependence on the container that a working coupling keeps invisible — outward effect was always by propagation from one's own subset, never a reach across a gap.*

15 First-class higher-frame quantities

Proposition 15.1 (Higher-frame first-class structure)**.** *A quantity concerning G may be absent from G while existing as ordinary first-class state in containing frame W that represents the relation from which the quantity is defined.*

This is the basis on which later papers introduce Narrative, standing, ownership, and other relational quantities without bloating the primitive graph.

16 Causal order and local time

Realised collapse induces an ordinal event order

$$e_0 \prec e_1 \prec e_2 \prec \cdots .$$

Cross-boundary interaction adds only represented causal relations.

Proposition 16.1 (Ordinal sufficiency)**.** *The substrate supplies causal order but requires no primitive real-valued metric duration.*

Metric time is a grounding, e.g.

$$\theta : \mathcal{H}_G \to \mathbb{R}, \qquad e_a \prec e_b \Rightarrow \theta(e_a) < \theta(e_b).$$

The separation between a primitive causal partial order and additional metric structure has a close analogue in causal-set approaches to spacetime (Bombelli et al., 1987). The present substrate does not assume causal-set dynamics; the citation marks the established precedent for keeping order and metric distinct.

The chained notation $e_0 \prec e_1 \prec \cdots$ is the *in-frame* sequence and should not be read as global. Rules require ordering — closure and cause-before-effect both presuppose it — but the requirement is a *partial* order, weaker than time.

Proposition 16.2 (Partial, not total; carried by the rules)**.** *Two collapses are ordered only where a boundary relates them; collapses not so related are unordered — concurrent, incomparable. The ordering is intrinsic to the rules, since one delta's being what enables another's match is precisely their precedence, so no external clock is needed to impose it. A single frame's own writes are totally ordered (its inferred time, Section 18); across frames only the partial order holds, and a total order is one frame's linearisation of it. The in-frame total order is not imposed but follows: change is self-change on one structure (Proposition 10.5), so two of a frame's own changes cannot be concurrent — the second is on the result of the first — and there is nowhere for an in-frame tie to sit. A tie needing resolution would be two collapses causally related yet unordered, and that never arises: causally related ones are self-change on each other's result (ordered), and unrelated ones are rightly unordered (concurrent, owing no order). The apparent in-frame concurrency that would have needed an arbitrary tie-break was an artefact of picturing changes as writes to a shared buffer; with change as self-change, it is absent.*

Time was accordingly a *total* order imposed where only a partial one is required; removing the substrate time dimension removes the over-ordering, not the precedence. This is the same conclusion reached, from engineering necessity, by distributed systems without a shared clock: the happens-before relation is a partial order with per-process sequence counters and no global instant (Lamport, 1978), and consistency across processes is an expensive coordinated property rather than the default (Brewer, 2000). The non-atomicity this implies — a graph's state may be altered by another write while its own rules are mid-run — is not a defect here but the expected regime; robustness is obtained by re-running to fixed point (a raced deformation becomes residual and is re-encountered on the next pass), not by locking.

Remark 16.3 (Globally asynchronous, locally synchronous)*.* The partial order permits both synchrony and asynchrony, and since only rules-running is compulsory (Remark 12.9), neither is required — that both occur is an observation about the groundings treated here and not an inference from their being optional. Where a boundary couples collapses, they are ordered together — a local synchronous domain, which is what a bound or driven subgraph is; where no boundary relates them, they are asynchronous. A subgraph may accordingly be *ticked* by a containing frame N levels above it (the driven mode of Proposition 16.4, a rule in a superframe advancing the subframe's succession) or run on its own change (the closed, self-looping mode). Synchrony is the coupled case and costs a coupling write; asynchrony is the uncoupled default. The substrate is therefore globally asynchronous and locally synchronous where coupling exists — the architecture distributed hardware knows as GALS (Chapiro, 1984) — and bound physical systems (an atom, an orbit, a lattice) are exactly its local synchronous domains, while causally disconnected regions are its asynchrony. Any earlier phrasing suggesting the substrate is *entirely* asynchronous should be read in this sharper form.

Proposition 16.4 (No intrinsic first or last; entry is chosen from the container)**.** *By Axiom* −1 *the event order has no intrinsic start and no intrinsic end: a first would complete the order backward and a last would complete it forward, and neither is available. A subgraph process that*

does not terminate within itself is therefore begun not from within — it has no first rule — but by a rule in a containing frame that chooses an entry point into the subgraph's open order.

This is why a grounding's initial and boundary conditions are supplied separately from its laws: the laws are the ordering, which fixes no start, so the entry must be imposed from outside, contingently. It is also the substrate description of booting a situated graph, and of the "no natural ending" that follows for any such order — one enters and leaves at chosen points, neither of which is intrinsic to the material.

17 The present, and the states of a dimension

Definition 17.1 (States of a dimension)**.** *For a frame $\mathcal{F}$, a dimension is in exactly one of four states:*

$$\boxed{\text{NULL} \neq \text{undetermined} \neq \text{undisclosed} \neq \text{resolved}.}$$

NULL*: the dimension is not instantiated; there is nothing to have a value.* Undetermined*: the dimension is instantiated and* no value has been realised — *collapse has not occurred.* Undisclosed*: a value has been realised but is not held by this frame.* Resolved*: a realised value, held.*

Remark 17.2 (Why the middle two must be separated)*.* Ordinary usage collapses three distinct conditions into the single word *unknown*, and only one of them is epistemic:

1. **undisclosed** — a value exists and this frame does not hold it. Another frame may. This is ignorance.
2. **undetermined** — *no value exists yet.* Nothing has been realised, so there is nothing to be ignorant of. This is not a deficiency of any frame.
3. **indeterministic** — no structure fixes what the value will be. The present theory denies this (Section 22); it is listed to be excluded.

The distinction is operational rather than terminological. **Determination is an event, not a fact about a pre-existing value**: "undisclosed" presupposes a value already realised and merely unheld, whereas "undetermined" asserts that collapse is what produces one. A frame may act to *obtain* an undisclosed value; it can only *await or participate in* the collapse that determines an undetermined one; and a NULL dimension is not something it can look for at all.

This separation is what Proposition 17.4 states from the other side: a continuation may be determinable and undetermined simultaneously, since the first concerns what fixes it and the second whether it has happened.

Definition 17.3 (The present, structurally)**.** *Within a frame, the present is the collapse boundary at which every instantiated* undetermined *dimension relevant to that collapse has been determined:*

$$\boxed{U_{\mathcal{F}} = 0.}$$

$U_{\mathcal{F}}$ counts undetermined dimensions only. NULL dimensions are excluded — they are not variables. ***Undisclosed dimensions are also excluded****: they are already determined, and a frame's failure to hold them is an epistemic deficit rather than an unresolved collapse. A frame with outstanding undisclosed structure is still at its own present.*

The present is therefore defined structurally rather than as a point on a primitive time axis, which is required given Section 18.

Proposition 17.4 (Determinable is not determined)**.** *A continuation may be completely determinable from present structure without having been realised:*

$$\boxed{\textit{determinable} \neq \textit{determined.}}$$

"Determined" is reserved for what has collapsed into realised history; a continuation that is determinable but not yet realised is undetermined *in the sense of Definition 17.1, and this is not a statement about any frame's knowledge.*

Proposition 17.5 (Causally sufficient present structure)**.** *The determining condition is structure instantiated* now*, including relevant containing-frame relations:*

$$\boxed{\textit{same causally sufficient present structure} \Rightarrow \textit{same realised continuation.}}$$

This replaces any formulation requiring a complete antecedent state *in time*. Only the present exists in a frame; past states need not remain extant, and the future is not an extant region lying ahead of the present. The future consists of reachable or determinable continuations represented now.

Corollary 17.6 (History principle, strengthened)**.**

$$\boxed{\textit{history matters only where it survives as present structure.}}$$

The past is surviving consequence instantiated now, not a stack of still-existing historical frames.

Present structure is therefore *required* — it is the antecedent the next transition reads — while a history *record*, a stack of past states, is *optional* and in no way required by the dynamics. The causally relevant past already lives in the present structure (which is what the corollary says), so forgetting the record loses nothing the dynamics use: a graph with no history at all runs identically, the substrate's strong Markov property that the continuation depends only on the present and never on the path (Propositions 17.5, 13.2). A record is accordingly a paid copy kept for the copier — audit, replay, proof — and never funded by the process. Anterograde amnesia is the clean case: present structure (procedures, skills, the operative past) works while record-forming is broken; function needs the effects, never the diary. This is the past-directed half of "the future is not an extant region ahead": neither past nor future is a required region the graph moves through — there is present structure, and optional copies.

18 Time is not a dimension of the substrate

The claim here is stronger than non-primitivity, and stating the weaker version invites the wrong reading. It is not that time is nonlinear, or differently shaped, or emergent-but-still-there. It is that

Axiom 0 (No substrate time dimension)**.** Time is not a first-class dimension of the substrate. *What the substrate has is* change*: realised collapses. It has no ratchet over and above them — no mechanism that manufactures succession — because the three properties usually bundled under that word are readings over changes or properties of them, not a further thing the substrate runs. The ordinal event order of Section 16 is carried by the changes' own enabling-relations; direction is the no-copy asymmetry of change, since a change does not preserve its predecessor; and persistence is only that further changes occur. A time dimension exists only where a frame instantiates one* to represent *that order.*

Remark 18.1 (Even succession is not a substrate mechanism)*.* This removes the last piece of machinery a reading of the earlier drafts might have kept. Succession, an ordered "ratchet"

turning under the changes, is to change what a metric time dimension is to succession — one level of over-structure, nameable but not required. Step the removal through: take away any posited succession-engine and the before/after is still present, carried by which change enables which; the arrow is still present, as change's own non-preservation of its predecessor; the running is still present, as further changes occurring where rules fire. Nothing is lost, so the engine was never load-bearing. The substrate has change and the absence of change, and nothing between them.

Proposition 18.2 (Time is observer machinery)**.** *For a frame $\mathcal{F}$,*

$$\boxed{T \in \mathcal{F} \iff \mathcal{F} \textit{ instantiates the relation representing succession.}}$$

Otherwise $T =$ NULL *in that frame: absent, not set to a value. A substrate frame that realises continuations without representing succession has no temporal dimension at all, and nothing about its dynamics is thereby incomplete.*

The distinction that carries the axiom is between *succession* and *time*. Succession is a relation among realised changes, carried by which change enables which; it is substrate-level in that sense, not a mechanism the substrate runs over and above the changes. Time is a represented dimension, and representing something requires a frame that represents it. Only an agent — in the minimal sense of a frame that instantiates the representing relation — observes time.

Schematically, and without pretending to explain the first arrow,

$$? \dashrightarrow G \xrightarrow{\text{realised change}} \text{changes, enabling-order carried} \xrightarrow{\text{a frame represents an order}} T \in \mathcal{F}.$$

The second arrow is substrate dynamics. The third is representation, and it is optional.

Proposition 18.3 (Time is inferred, not observed)**.** *No collapse presents a temporal coordinate. Succession is a relation* between *realised states, and a relation between states is not a datum of any one of them. A frame therefore acquires a time dimension only by* inferring *it from the relations among what it holds. Consequently*

$$\boxed{\textit{a frame observes time} \implies \textit{that frame is capable of inference.}}$$

Corollary 18.4 (Reactive frames have succession and no time)**.** *A frame that transforms under the Graph Law without inferring relations among its held states participates fully in realised change and has $T =$* NULL*. Nothing in its dynamics is thereby impaired: change is what the substrate supplies and the order among changes is carried by them, while time is what an inferring frame constructs from that order.*

Remark 18.5 (A rate has no in-frame denominator). A realised delta is a difference, not a difference *per* anything: what a frame holds is Δ, never $\Delta/\Delta T$, because ΔT is the inferred dimension of Proposition 18.3 and is not available in-frame to divide by. A rate is thus a second construction — an observer dividing an accumulated Δ by its own count of writes — and is cross-frame in the same sense that a metric duration is. This is why a substrate law must be stated in Δ rather than in a rate: "constant motion" is the standing structure persisting without a write (Remark 6.3), its momentum quantity a difference Δ and not a velocity. It also diagnoses a familiar regress in constructed learners, where every adaptive rate is found to need a reference rate without end: the regress is the signature of placing a cross-frame quantity in-frame, and it disappears when the held quantity is the difference itself.

This orders two capabilities that are usually treated as independent. Inference is not a faculty a temporal agent happens to have; it is the precondition for being a temporal agent at all. It also grades: the temporal structure a frame holds is bounded by the inferential structure it can construct, so "how much time" a frame has is a question about its inference and not about the world.

Remark 18.6 (Empirical instance). This is measurable rather than merely stipulative, and it has been measured. A perceptual system given ordered frames and no temporal coordinate infers a transition generator carrying one frame to the next, and the integer count of applications *is* the time axis it then holds; given the same frames shuffled, it infers none. The dimension was not supplied to it and does not exist in the observations — it exists in the map between them, once something constructs it. Notably, the mechanism that constructs it is the system's general *dimension-inference* machinery, not a temporal faculty: time is minted by the same operation that mints any other unobserved dimension, which is Proposition 18.3 in implementation.

Proposition 18.7 (The unit of inferred time is a cycle, and requires closure)**.** *A frame does not count its individual changes — those are below its aperture — but its own* cycles*, each one traversal of a closed rule-loop. The unit of inferred time is therefore a cycle, not a change, and a cycle has no fixed length: each does a variable amount of change, so inferred time is non-uniform in underlying change (a nonlinear grounding, which Discrete Sufficiency already permits) and a fuller cycle reads as a longer moment (subjective dilation). A cycle exists only for a graph with* closure *— a graph without it runs to fixed point once and stops, one pass and no repeated tick — so having time requires closure as well as inference (Proposition 18.3): a one-shot computation, lacking a loop to iterate, has no time, and a frame's experienced "now" is one cycle of its own loop.*

Remark 18.8 (The processor is a constructible demonstration). A running program is a worked instance, carried by most people on earth. An instruction is the program's step, and it does not take a fixed physical time: an addition is one cycle, a memory load one on a cache hit and hundreds on a miss, a branch may stall, and the core reorders and overlaps beneath it. Yet from the program's point of view every instruction is *one* step — its own unit, read as 1 by construction, since a step is a step — so the program's inferred time is its step-count, uniform to it and non-uniform in physical time. The variable timing's *mechanism* is opaque to the program — the cache miss and the reorder are below its aperture, beneath its primitives (Proposition 19.4) — while its *effects* are not, since a program can time its own reads and infer what it cannot see, which is the leak of Proposition 19.2 and the basis of the attacks cited there, which is why "time is a variable-length cycle, not a fixed duration" reads as exotic even though the device demonstrates it continuously. How much weight this carries requires separating two things, and the separation recovers something real.

The *vocabulary* is borrowed and carries no weight whatever. Aperture, runner, primitive, cost: this account took them from engineering (Section 23), so terminological agreement between the two is not agreement about anything, and a reader who notices that this reads like computing has noticed only that.

What is not borrowed is that computing was *forced* to these arrangements. The right question is not why this account looks like computers but why computers look like this — and the answer, case by case, is that engineers fought each of these features and lost. Nobody wanted a memory hierarchy; it exists because latency and bandwidth do not scale together. Nobody wanted instructions of unequal duration; they exist because operations genuinely cost differently. Nobody wanted opacity between levels; it exists because complexity must be bounded to be buildable at all. And nobody wanted to give up a global clock: distributed systems have no shared instant because signal speed is finite, which is a theorem rather than a preference (Lamport, 1978), and the resulting architecture is a concession to physics that engineering resisted for as long as it could (Chapiro, 1984; Brewer, 2000).

So the convergence is not that two descriptions sound alike. It is that a discipline optimising hard against cost, with no interest in metaphysics and every incentive to avoid each of these features, arrived at cycle-as-variable-unit, per-frame logical time, per-level opacity and cost inherited along provision — because the constraints that force them are the constraints this account claims are the operative ones. That is evidence about the constraints, and it survives the borrowed vocabulary because it does not run through it.

The honest limit is that computing's designers are people, so convergence could reflect shared cognition rather than shared constraint. What answers that is the forcing arguments themselves being physical — finite signal speed, finite bandwidth, a thermodynamic floor — rather than matters of taste, and Lamport's result in particular being a proof about what is available given a finite c, not a design that could have gone otherwise.

Remark 18.9 (Physics groundings are developed in a companion). The reading of c as a frame's own change-throughput unit, and the shapes that follow from it — the invariance of c as a theorem, the reconciliation of special and general relativity as one change-budget spent on motion or on change-load, the dimensional constants as frame-units with the uncertainty principle and the Landauer floor among them, the Planck scale as the frontier, and the second law as no-copy accumulation — are developed in the companion paper, *Graphic Equalisation — Physics Grounding*. They are information-level groundings in the sense of Section 1, and are kept out of the present paper so that the substrate theory stands without them.

19 Primitives, and the closure of their provenance

The provenance of r (Section 1) records that operative transformation requires something that performs it. What performs it is real; naming it as a thing is the error this section avoids, because a graph never meets it. What a graph meets is its *primitives*.

Definition 19.1 (Primitives)**.** *A graph's* primitives *are the operations it invokes without performing: the operator set its rules bottom out in. Every level has them, and having them is what being a level consists of — a graph with no primitives would have to perform everything it invokes, which is the regress this substrate does not contain. That is the whole of the definition, and in particular it says nothing about what a graph can learn of where they come from.*

Two different things can be meant by provenance being hidden, and running them together inverts the relation between them.

Deliberate exposure — *state maintained so that what runs on it can read it, as reflection does — is structure doing work for something other than the frame holding it, and is therefore excluded by well-formedness (Definition 12.5) in the ordinary way: it costs, and the frame supplying it gets nothing.*

Leakage *is the opposite case and is not excluded at all. A cache earns its retention many times over and leaks precisely* because *it does work; an arithmetic fast enough to overflow at a word boundary discloses the boundary by being fast; a page-sized performance cliff exists because the page size is doing something. Every leak in Proposition 19.2 is a consequence of efficient design rather than of poor design, so well-formedness does not prevent leakage — it* produces *it, and a structure that leaked nothing would have given up the optimisations that distinguish it from a uniform one, which is what constant-time implementations do and what they pay for.*

Proposition 19.2 (Primitives leak, and complete provenance is still unrecoverable)**.** *Leaking is ordinary rather than exceptional. A primitive's timing discloses the structure beneath it, which is why measuring how long a read takes recovers what a cache holds (Osvik et al., 2006; Kocher et al., 2019); an arithmetic that overflows at a particular magnitude discloses a word size; a rounding discloses a representation; and a performance cliff at a particular size discloses a page. Most abstractions leak, and this one is not special.*

How a leak reaches a frame at all is worth stating, since what falls outside an aperture is absent *(Definition 12.2) and an absent thing cannot disclose anything. It does not arrive by perception. What arrives is an ordinary in-frame observable on a dimension the frame already instantiates — a duration in its own units, a value at a boundary — and what the frame does with it is* infer*. The cache is not perceived; the timings are, and the cache is concluded from them.*

What does not follow is that provenance can be recovered entire. Every act of recovery is itself performed by primitives, so a graph inferring what lies beneath one operator does so using others whose own provenance is not thereby in view, and there is no position from which the whole supply is inspected at once. The result is the standing shape: partial always, complete never *— leaks give a graph real and sometimes decisive knowledge of what supplies it, and never the closure of that question.*

This is what the physics companion's treatment of c turns on rather than contradicts. A frame cannot read the rate beneath its primitives directly, and has nonetheless obtained a number for it by measurement and inference across three centuries — which is exploiting a leak, and is available precisely because provenance is not sealed.

Proposition 19.3 (Primitives are supplied, not intrinsic)**.** *A graph is data — nodes, relations, and rules — and the operations its rules bottom out in are supplied to it rather than contained in it. The same graph may therefore be run against different supplies of the same operator set, and nothing supplying an operator need be the substrate the graph describes or is "in." Where the work is done is a fact about that supply and not about the graph.*

At one extreme the coupling is tight — the graph is fused to what supplies it, as matter running its own change or a mind running on the tissue that is it — and the compute is local, in the graph, inseparable. At the other it is loose — the graph is data executed by a separate interpreter — and the compute is done in the runner, elsewhere, while the graph becomes portable: copyable, relocatable, and run at whatever rate the runner supplies (Section 18). This separability is the condition for the copyable-generator and clock-independence results (Proposition 12.11, and the Seldon paper's dependent-clock remark): a graph welded to what supplies it has none of them, a graph loosely coupled has all of them.

Proposition 19.4 (A well-formed runner does not expose itself, and no level is self-transparent)**.** *The asynchronous, locally synchronous structure a frame observes (Remark 16.3) is the graph's* internal *view and does not fix its runner's actual mode. A hidden central runner may execute a graph serially, one change at a time, while the graph — lacking any access to the runner's clock — appears asynchronous and concurrent to itself. More generally, where a frame has no boundary to the substrate that runs it, its runner lies beyond the frame's epistemic boundary in exactly the sense of Section 1.*

Two claims have to be separated here, and an earlier statement of this proposition ran them together by calling the runner unseeable rather than unseen. That a runner is hidden is contingent*. Nothing forbids a runner from exposing its own state to what it runs, and computational reflection does exactly that, as this paper acknowledges when placing itself against that literature (Section 23); a tower in which each level is available to the one above is a built artefact, not a thought experiment. A runner that does not expose itself is the* well-formed *case, and for the reasons well-formedness always turns on: exposed runner state is structure maintained for work the frame does not do, and each exposure needs a runner of its own to run the reading of it.*

What is not contingent is weaker and survives reflection: no level is self-transparent*. Reading is an operation whose product lands one level up (Remark 12.7), so a level cannot hold the reading of itself while being the level that is running. Reflection makes the levels explicit and pays for each; it does not terminate in a level transparent to itself. A frame may therefore be given sight of what runs it, and cannot be given sight of its own running. Since the rules are graph and a frame can read and rewrite them (Definition 10.2), while its own running is not graph it holds, the asymmetry is exact: a frame is transparent to itself in precisely one respect and opaque in precisely the other.* It can know what it is and cannot observe its being it. *Self-modification is available; self-observation at the level of the modifying is not.*

This is constructively demonstrable rather than merely argued. A computation realised inside another rule system — a graph whose rules are implemented within the mechanics of a game or a universal cellular automaton (Berlekamp et al., 1982) — is a blind subframe: from inside it computes at its own change-count and has no means to detect that its "compute" is the tokens of the host, or that the host may be advancing it by a single serial loop. Building one confirms the opacity directly, and puts the observer of any frame on the same footing. Consequently the question "where is the universe's compute done" has the precise form: it asks after the provenance of the universe's own primitives — self-supplied, or supplied from a substrate not itself — and is undetermined from inside. The theory declines to decide it; it is a question about a coupling below the frontier, and the demonstrable opacity is why no internal observation settles it.

Proposition 19.5 (A primitive is another graph's running, exposed)**.** *Nothing supplying a primitive is a substance apart from graph data. An operator is the observable running of some other graph, which has primitives of its own, supplied in the same way. A graph's "primitives" are therefore the observable running of other graphs, exposed as an operator set, with no primitive runner at the base. A single chain — code over machine code over microcode over silicon over electromagnetism over quantum physics — is one path through that provision, and a common one, but not its shape: a process takes arithmetic from its processor, file operations from a kernel that relays them to a device, and remote calls from a machine that is not beneath it in any sense. Fan-out and relay are ordinary, and the tower is the degenerate case in which provision happens to be a chain. The regress bottoms not at a floor but at the epistemic frontier of Section 1: below the lowest measured level is the unclaimed pre-relational substrate.*

Two consequences. Opacity is per *operator* rather than per level (Proposition 19.4): a graph uses its operators and cannot see beneath any of them, so its operator set is its frontier — and since provision may fan out, it cannot tell which of its operators share a provider and which do not. Two primitives supplied by one supergraph and two supplied by different ones are in-frame indistinguishable, so a graph does not know how many runners it has, or that the number is one. And economics is inherited along provision: an operator costs what the runner of that operator charges, which costs what *its* providers charge, down to whatever floors each route reaches (Remark 12.6). Where provision is a chain this is one bill passed upward, and the cost of a thought is the biological cost is the biochemical cost is the electromagnetic cost; where it is not, it is a sum over provenances whose routes may differ in length and may not all terminate at the same floor. The same form, graph data that runs, sits at every level: each level *is* the structure, because a runner is the same kind of thing it runs.

A rule's action, moreover, is a *pointer* into its runner, not a copy of the work. The work is done once, in the runner, and every rule invoking that operator points to the same doing — like a shared operation defined once and called from everywhere, never re-implemented. Two things follow. Work is never duplicated, which is why copying a generator copies its pointers and shares the pointed-to work: the copy costs only its addressing, not its labour (the economic root of the copyable-generator result, Proposition 12.11). And a rule's rate is not a property of the rule but the rate of the operation its pointer lands on — a shallow, cheap tier or a deep, slow one. Rules take different rates because they point to work done at different depths of the stack, and the rate is inherited from there rather than assigned as a knob on the rule: to make a rule faster one points its action at faster work, never tunes a rate parameter it does not have.

20 The single prohibition, and its formal core

The account forbids very little, and what it does forbid is one thing recurring rather than a list. Stating both facts together is necessary because each is misleading alone.

The permissiveness first. Time is not compulsory; a block reading is not prohibited; one-dimensional time is not forced; a globally shared counter is not forbidden, only expensive; and in the physics companion, complex amplitudes are not forced and the one-way speed of light is not unmeasurable *simpliciter*. A framework that forbids little predicts little, and from outside this is exactly what unfalsifiability looks like. The discriminating question is not how many impossibilities an account holds but where it puts them: this one puts almost none at the substrate and a great many in the groundings, and says in advance which is which. That is deliberate and it has a cost, which should be stated rather than defended — the substrate risks little, and what risk this work carries is carried by the groundings. A reviewer who objects that the substrate forbids too little to be tested is right about the substrate.

What it forbids is that *a frame cannot get outside itself*. Concretely, and confined to what is actually derivable here: a frame cannot be transparent to itself at the level it is running at (Proposition 19.4); it cannot measure its own throughput in units independent of it, since the measurement is funded by the throughput — which leaves a number obtainable and its absolute scale not; and it cannot hold a view from nowhere, which is Axiom −1. Two further candidates are deliberately *not* on this list. That a frame cannot obtain a quantity whose unit does not close within it is contingent rule content and the physics companion removes it explicitly. That a frame cannot decide anything without writing is the write-economics, which Remark 12.10 carries as a pattern rather than a universal. Folding either into the single prohibition would be the over-unification this section otherwise warns against.

This is not a fourth answer to what is compulsory, and the paper should not read as though it were. Remark 12.9 says only that rules run is compulsory; rules running requires a relation across which they run, so the container relation is not an addition to that but its content. The prohibition is not a further necessity alongside them: it is a constraint that holds wherever rules run at all, which is why it is not rule content and cannot be varied by changing the rules.

The formal analogue

The standing of that prohibition is unlike the rest of this paper, and the difference should be stated exactly, including where it falls short. The rest of what is claimed here is contingent rule content. The impossibility of a level being transparent to itself has a *formal analogue*, and the analogue is proved — in mathematics rather than in physics, so it does not depend on any fact about which universe this is.

A consistent, effectively axiomatised formal system with enough arithmetic strength — from which the expressibility of its own syntax follows, by numbering, rather than being assumed — does not prove its own *Gödel sentence*, which is nonetheless true in the standard model of arithmetic (Gödel, 1931). A language of sufficient strength cannot define its own truth predicate under classical bivalent semantics; truth for it must be given in a metalanguage (Tarski, 1936). And no single procedure decides halting for arbitrary procedures — a limit on what a universal procedure can decide, and one that arises *because* such a procedure exists rather than in spite of it (Turing, 1936). These are not three coincidences: they share one diagonal construction, and the categorical statement of it makes the unity explicit (Lawvere, 1969; Yanofsky, 2003). Tarski's is the tightest fit — the semantics of a level is available only from above it, which is Proposition 19.4's surviving half in the vocabulary of formal languages.

Four qualifications, none of them optional.

The theorems carry hypotheses that must be carried with them: effective axiomatisability and sufficient strength for the first, classical bivalence for the second — a language may contain its own truth predicate at the cost of bivalence (Kripke, 1975) — and, for all of them, that a system be strong enough to represent its own operation before any of it bites. A frame too weak to model itself has no self-transparency problem, because it has no self-model.

The unification covers what is claimed above and not more. Gödel's *second* incompleteness theorem — that such a system cannot prove its own consistency — is the more striking result and

is not covered by the unification as published: Lawvere (1969) closes by noting that extending to it may require a specific relation between formulas and their numbers that his framework does not supply, while Yanofsky (2003) treats it as a simple consequence of the first and expects the scheme to reach it. The claim here is made with the first theorem, which the unification does cover, and takes no position on the second.

The mapping from frames to formal systems is an argument in this paper and not itself a theorem. What is proved is proved about the formal systems. The honest statement is therefore that the impossibility has a formal analogue whose analogue is proved, and that the mapping is argued — which is weaker than "the prohibition is proved" and is the strongest standing anything here has.

And the substrate-internal derivation of the same conclusion, by way of a read being an operation whose product lands one level up (Remark 12.7), inherits the write-economics' contingency and does not establish a necessity. Remark 12.10 carries that economics as a pattern open to counterexample, and §24 stakes a falsifier on it. The formal analogue is what carries the weight here precisely because it does not depend on the economics; an earlier version of this argument rested the necessity on the internal derivation alone, which cannot bear it.

What the results forbid is *complete* self-capture and not partial self-modelling, which is why reflective towers work, arithmetic is not paralysed, and a frame can know a great deal about itself. Partial always, complete never — and the asymptote in that is not a figure of speech, since a diagonal argument does not say a target is far away but that for any level reached there is another above it, and none is the last.

21 Carvings, and why classification degrades

Since the rules draw the boundary (Definition 10.2) and the rules are contingent, frames are *arbitrary*, and the objection that follows should be met rather than deferred: a notion satisfied by any cut explains nothing.

The first half is granted and is the point. No carving is privileged, nothing in the substrate marks which subgraphs are the real ones, and this is Axiom −1 reaching the last place a privileged view could still have been hiding — not in the observer, not in what it observes, but in what counts as an observer at all. A rock is a frame, an imagining is a frame, a population is a frame, and none of these is an inclusiveness conceded for breadth.

The second half does not follow, because arbitrary is not unconstrained. The rule set must be admissible; a frame whose rules never close a cycle has no unit, no inferred time and nothing to observe with; and well-formedness is graded (Definition 12.5), so a badly drawn frame is expensive, incoherent or blind while remaining a frame. Any cut is a frame and nearly every cut is a poor one — the same shape as the rest of this account, in which almost nothing is forbidden and the differences are prices.

The mechanism of that price is worth stating exactly, because the obvious statement of it is wrong under this paper's own economics. Holding a classification is not itself expensive: Proposition 12.4 makes the standing structure's persistence free. What an ill-formed cut costs is *writes*. A cut that does not fit what arrives forces its rules to write against the standing structure repeatedly, where a well-formed one rarely needs to — which is the observation already recorded after that proposition. And no selection principle follows from this: nothing in the substrate removes expensive structure. What follows is weaker and sufficient, that a frame under a budget will tend to be running cheaper cuts than dearer ones, other things equal.

Proposition 21.1 (Convergence is observed, not enforced)**.** *Two frames doing the same work converge on similar cuts as each becomes better formed, because the cheap cut for a given task is nearly unique. This is a tendency of cost and not the discovery of a privileged boundary, and the two are distinguishable in principle: a privileged boundary would* enforce *convergence, and*

enforcement would extend to the hard cases; cost-shaped cuts converge where the work is shared and come apart where it is not. The step from shared work to similar cuts assumes the cheap cut for a task is nearly unique, which is an empirical posit and not a result of this account; it is labelled again where it is used (Proposition 21.2).

Two further consequences follow from that proposition, and they belong here rather than where an earlier draft put them — nine pages upstream, inside a remark about processors, deriving a result from a premise the reader had not yet met.

Proposition 21.2 (Convergence under compression)**.** *A structure operating under a constraint, and under pressure to spend nothing on what does no work, approaches the arrangements that constraint forces — asymptotically, and only over the region the constraint touches. Two such structures therefore come to resemble each other without either having access to the other, because both are downstream of the same forcing rather than converging on a target.*

Three things about that statement are load-bearing.

It names no privileged arrangement. An earlier version had structures approaching *the arrangements a universe already uses*, which reintroduces exactly the privileged carving Proposition 21.1 denies, and additionally attributes work and purpose to a universe, which this account does not do and which that proposition's antecedent requires. What converges is the solution a constraint forces; a universe's arrangements are one thing that constraint has shaped, not a destination anything aims at.

The antecedent is about *expenditure* and not about contents. Stating it the other way — as pressure to hold nothing idle — would make the proposition false of every case it is applied to (Definition 12.5).

It rests on a posit that should be labelled rather than folded in. The step from *under a constraint* to *approaches the same arrangement* needs the cheap solution under a given constraint to be nearly unique — and that is an empirical claim, not a result of this account. It is not derived here and it is not derived in Proposition 21.1 either, which asserts it in the same words. What can be said without it is weaker: a binding constraint narrows the solution space, and the more binding it is the narrower it gets. Everything below rests on the weak form.

And the rate is set by how binding the pressure is. A domain with slack to spend on anything but cost converges slowly or not at all; one optimising hard against cost per unit of function converges fast. Computing is the second kind, which is why it is the worked instance and why decades sufficed.

Remark 21.3 (Evolution under the same treatment: independent where computing is not, and weaker where computing is strong)*.* Descent with modification is compressed in the required sense — energy, time and reproduction are all scarce, and the pressure falls on what is *spent* rather than on what is carried.

Whether that makes it a better instance than computing depends on which axis is asked, and an earlier version of this remark asserted superiority on both, which is not available. On *independence* it is decisively better and this is the axis the argument needs: computing shares this account's vocabulary and is disqualified for it, whereas biology shares neither vocabulary nor personnel, and its convergences were catalogued long before anything here was written. On the *antecedent* the ranking reverses. The rate law above sets convergence by how binding the pressure is, and computing was named the hard-optimising kind for a reason; biology is the slack kind, since selection is efficacious only where its coefficient exceeds the reciprocal of effective population size, so structure whose cost falls below that line is invisible to it and persists (Ohta, 1973). Biology is therefore a *sufficient* instance on the antecedent rather than a superior one, and the paragraph that follows is the work of establishing that sufficiency rather than a defence of a stronger claim.

An objection is worth meeting here rather than leaving to a reader, because it looks decisive and is not. Genomes conspicuously carry structure that does no work — pseudogenes, transposon

debris, whole duplicated regions — which appears to fail the antecedent outright. It does not, because the antecedent is about spending. *Inert sequence is the cost of its own deletion, represented.* Carrying it costs replication, and it costs search — more sequence to traverse, more places a regulatory element could be, more targets for recombination — but each of those is small and continuous, while removing it is a single write that risks excising something that turns out to matter. So it persists wherever the accumulated search cost has not yet overtaken the cost of the write, which is most of the time. Constructive neutral evolution is that observation made precisely rather than a counterexample to it (Stoltzfus, 1999; Lukeš et al., 2011).

The concession also earns something, because the weakness just admitted makes a quantitative prediction the account would otherwise not get. If inert sequence persists exactly where its cost sits below what selection can resolve, then how much of it a lineage carries should track that threshold — less in large populations where selection is efficacious, more in small ones where it is not. That inverse relation between genome bloat and effective population size is observed, and has been argued to be the explanation of genome size itself (Lynch and Conery, 2003).

The reading is contested (Whitney and Garland, 2010), and that is the expected condition rather than a weakness in it. What the economics predicts is a *trend*, and it is not entitled to more. The three costs vary independently across lineages, and so do mutation rate, deletion bias, recombination and transposon activity, so nothing here licenses a clean relation — and a clean one would indicate a single dominant term the account does not posit. The exposure is accordingly to *direction* and not to tightness: a phylogenetically corrected null leaves the account where it was, whereas bloat tracking effective population size *positively* would contradict it outright. That is a weak commitment and is stated at its strength. It is also the same commitment made twice already — convergence is observed and not enforced (Proposition 21.1), and cheapness is a tendency of cost rather than the discovery of a joint — so reading a trend as a trend is this account's standing position and not an accommodation made once the evidence turned out noisy.

The same shape appears wherever the same economics does, which is the point rather than an aside. Dead code survives in working systems for the identical reason, and shows the later costs plainly because people pay them. Leaving it is nearly free per line; the accumulated search cost is that everything is harder to find; and the cost that forces the issue is a change made carefully to a path nothing executes — paid in full, returning nothing, and leaving its author believing the system was altered. Repair machinery maintaining non-functional sequence is the same expenditure one level down. Neither is certain waste, since inert structure is sometimes being prepared for reuse and dead paths are sometimes revived, so what grows with the held quantity is the *expected* share of effort that lands on nothing.

That last hedge is not a formality, and the case that makes it concrete also explains why removal is expensive in a way the preceding paragraphs only asserted. The vermiform appendix was classified as vestigial for over a century, on Darwin's own reading, and removed routinely on that basis; it is now argued to serve as a reservoir from which gut flora can repopulate the colon after the contents are lost (Bollinger et al., 2007), and to have arisen independently many times across mammals, which is not what a functionless remnant does (Smith et al., 2013).

The lesson is not that the appendix turned out useful. It is that *inertness is a judgement made at an aperture, and the judgement fails.* Something was called structure-that-does-nothing by people looking carefully, for a hundred years, and they were looking at the wrong scale — the function is not the organ's own activity but its relation to a population it houses.

Which sharpens why inert structure is retained, beyond removal being a write. Establishing that structure is inert requires perturbing it and observing that nothing follows — and that perturbation *is* the removal one was trying to justify. The information needed to license the write is obtainable only by making it. So the choice is not between a cheap test and an expensive change; it is between paying for the change without the test, or not paying. Under that structure, retention is the ordinary outcome, and it is a correct one rather than a failure of nerve. Neither

case is a hoarding failure. Both are what an economics in which holding is free and changing is paid predicts, and a lineage or a codebase that pruned aggressively would be the anomaly requiring explanation.

Convergent evolution is Proposition 21.2 with a name biology gave it first. Independent lineages arrive at the camera eye, the streamlined body, the wing, and they do so because optics, fluid dynamics and lift are the same wherever they are met, and the catalogue is large (McGhee, 2011). In the cases named here the forcing constraint can be stated, which is what this account requires of a convergence before it counts; that it can be stated in *every* case is not claimed, and how far convergence indicates inevitability is itself disputed (Losos, 2017). Nothing was copied and nothing had access to anything.

The change side is the same economics read the other way, and it gives a second correspondence. In this account change is paid and persistence is free, so a structure changes when what it meets will not cohere with what it holds and not otherwise — which is stasis as the default and change as the forced exception, the shape the fossil record was read as showing when long stable intervals were argued to be punctuated by short rapid ones (Eldredge and Gould, 1972). Read in the substrate's terms, selection is the residual gate: variation is undirected and cheap, and what is expensive is the rewriting that follows when the residual will not fit.

Grain is set by the environment. How finely a lineage discriminates is not chosen by it but forced by what it meets, which is the aperture claim of Section 12 in another grounding, and it is why the same organism is coarse in one setting and finely differentiated in another.

Remark 21.4 (A popular understanding equalises, and that is why it drifts). The evolutionary claims above read very differently depending on who is reading them, and the difference is not one of expertise so much as of aperture — which makes it an instance of the same machinery rather than an aside about communication.

The account of evolution most people hold is close to Darwin's: variation, competition, survival of the better fitted, over long stretches. Working biology in 2026 is not that. Developmental bias, phenotypic plasticity, inheritance beyond the gene, horizontal transfer, symbiosis and constructive neutral processes are all ordinary parts of the working picture, and the argument the previous remark takes a side in is an argument *within* that picture rather than an attack on it. So *natural selection is half of evolution* is heterodox against one aperture and unremarkable against the other, and a reader's reaction to it says which one they are holding.

Why the gap persists is the interesting part, and it is not that the explaining has been done badly. A popular understanding is a structure under compression like any other, and Darwin-level evolution is *well-formed for the purposes most people put it to*: it accounts for enough, it is cheap to hold, and almost nothing anyone meets in a day produces a residual it cannot absorb. It has equalised, and it persists for exactly the reason the horseshoe crab's form does — nothing is forcing a write. The working picture is the overshooting case, kept changing by a residual the field manufactures deliberately.

That predicts something checkable and slightly counterintuitive: the gap between popular and expert understanding of a field should track *daily exposure to residual* rather than education or communication effort. Where a lay frame is corrected constantly by contact — weather forecasting, where people check a prediction and see it fail — the gap is small and the lay frame keeps up. Where it is never contacted in a way that fails, the gap widens no matter how much explaining is done, because explanation supplies no residual: the frame is not wrong for what it is used for, so nothing in it has to give.

Two ends. This is not a claim that the popular account is ignorance — it is well-formed for its work, which is what makes it durable, and calling it deficient would be measuring it against work it is not doing. And expert consensus is not thereby correct; it is a frame under different pressure, which is a fact about its residual and not a warrant.

For this paper the practical consequence is small and worth stating: the evolutionary material is addressed to the working aperture, and a reader who finds it radical is invited to check it

against the field rather than against what they were taught.

Remark 21.5 (Overshoot is a rate, not a shape — and it is where agency comes from). Equalisation is a balance point: the state at which what a structure meets no longer forces it to write against what it holds. Reaching it is not the only thing a lineage can do at it, and the other option is the one that matters.

A lineage *overshoots* when its rate of change carries it past that point — when it goes on changing after the environment has stopped requiring it to. That is not a defect and not specialisation, which an earlier version of this remark wrongly made it. It is momentum in the rate itself: the machinery that produced change does not stop because the residual did, and a standing structure that includes a rate carries the rate forward, which is the subject of the Narrative Momentum companion one grounding over.

What overshoot produces is *capacity in excess of requirement*, and excess capacity is spendable in three ways of increasing consequence. It can fit an environment not yet met — structure arising under one demand turning out to serve another (Gould and Vrba, 1982). It can be spent on the current environment rather than on the organism, which is an organism altering the conditions it is selected by (Odling-Smee et al., 2003). And it can be spent selecting which environment to be in, which is also already named: *niche choice*, one of three modes in a published framework whose other two are construction and conformance (Trappes et al., 2022). The three-way division above is close to theirs and descends from the same source.

What this account adds is only the ordering — that the three are one capacity spent at increasing remove from the organism — and the suggestion that at sufficient accumulation the third amounts to self-direction. That last step is the one not found in the literature, and it should be taken carefully: the agency-in-biology programme grounds agency in organisational closure and normativity rather than in surplus, and there is a recent and pointed argument that the agency perspective has no research programme at all (DiFrisco and Gawne, 2025). Constituting agency out of unspent capacity is a position in a live dispute, not a summary of one.

Two ends, and the second matters more. Overshoot is expensive: excess capacity is structure that does no work yet, so it is ill-formed until the day it is not, and most of it is never spent. Most overshoot is waste, and most lineages that overshoot go extinct rather than becoming anything. And nothing here is directional — overshoot is undirected excess that is sometimes usable afterwards, and agency is one thing accumulated usable excess can amount to, not a destination the process is heading for. Read as a trajectory this would be orthogenesis, which the account has no room for, since the substrate supplies no goals for anything to be heading toward.

It is tempting to read that as two stable answers to a stable environment — settle at the balance point, or carry rate past it — and the temptation should be resisted, because the first is not available. *Equilibrium is a limit and not a state.* Reaching it would require an environment that stopped, and environments do not: an environment is other frames, which are running; a frame is a subset of its container, so its own persistence is a fact in that container; and at any grain where anything is individuated something is arriving. Total absence of forcing would require the container to be doing nothing, which is a limit rather than a condition environments occupy.

That is deliberately not identified with the empty limit this account reaches for perfection and for the view from nowhere, though an earlier version of this passage did identify them. Those are properties of a frame; equilibrium is a relation between a frame and its container. The shapes may be related and the relation is not argued here.

So what a stable-looking environment offers is not settling but a *rate of approach*, and the difference between the two answers is how fast a lineage closes on a target that keeps moving. One closes slowly and spends little; the other carries rate past the point the environment is currently requiring, pays for capacity that does nothing yet, and holds the means to survive

a change of conditions or to cause one. Neither arrives. The horseshoe crab is usually offered as the settled case, and the specialist measurements say otherwise — which is a correction that strengthens the reading above rather than embarrassing it, since under it nothing settles. The four-hundred-and-fifty-million-year figure belongs to the *lineage*; measured shape stasis in xiphosurids is Late Jurassic, on the order of a hundred and fifty million years, with developmental stasis older and body size increasing throughout (Bicknell et al., 2022). And the clade did not settle: xiphosurids invaded non-marine environments independently at least five times and twice radiated into novel morphospace (Lamsdell, 2016).

So the group is not an example of equilibrium attained. It is an example of equilibrium *approached slowly and never reached*: long intervals in which environments rarely required a write, a form that kept changing anyway, and repeated episodes of exactly the excess-spending the other rate describes. That is the prediction rather than an exception to it, and the received version — four hundred and fifty million years of a fixed shape — is the one the account would have had trouble with. What did continue underneath was demonstrated for these animals as a *decoupling* — large numbers of molecular characters distinguish even these morphologically conservative organisms, though the same study measures a moderate slowdown in sequence evolution rather than an ordinary rate (Avise et al., 1994) — and is now a stronger claim than point substitution: three whole-genome duplications are specific to the group (Nong et al., 2021; Castellano et al., 2025).

The other answer has a worked instance, and it is one where the received story was wrong in exactly the way this account predicts. Fish did not grow legs in order to walk. Limbs with digits appear in animals that were still aquatic and still breathing through gills: *Acanthostega* had eight digits on a limb (Coates and Clack, 1990) and retained internal gills (Coates and Clack, 1991; Clack, 2012), and digits and putative digits are now described inside a fin that still bore lepidotrichia, in an animal that still bore lepidotrichia (Cloutier et al., 2020). The structure preceded its *effective* terrestrial use — not its possibility, since the capacity for terrestrial locomotion is argued to have arrived with limbs themselves (Dickson et al., 2021), and the known *Acanthostega* material is juvenile and exclusively aquatic (Sanchez et al., 2016), so nothing here is claimed about the adult animal. Legs were not needed. What the lineage had was *change* — rate carried forward in its standing structure — and legs are what that change produced. Land became available to them afterwards.

And the sequence was not a ramp, which is the part that matters most for the reading offered here. *Qikiqtania wakei*, a Late Devonian elpistostegalian, has a pectoral fin specialised for swimming and lacking substrate-support features, which its describers read as a secondary return to open water close to the origin of limbed vertebrates (Stewart et al., 2022). A lineage near that origin specialised back toward the water — not, on this evidence, one shown to have held substrate-support capacity and then lost it, since the taxon sits more basally than *Tiktaalik* in the same analysis. That is the disclaimer above made concrete — most overshoot is not spent on anything — and it is the clearest available evidence against reading any of this as a direction the process was heading in.

What that instance is about is the half of the process *before* selection, and saying so places this account rather than merely correcting a story. *Natural selection is half of evolution.* Variation is generated and variants are differentially retained; neither half alone produces anything, and there is nothing to select from until something has been generated.

That is a claim about billing rather than about power. Selection's evidential standing is not in question and nothing here reduces it. What is in question is that the theory is named after one half while the other enters as a premise — *given heritable variation* — so that the structure of what is generated is treated as an input to explanation rather than part of it. If generation has structure of its own, the shape of what evolves depends on it as much as on what is retained, which is the substance of a live dispute rather than a neutral observation: an extended synthesis arguing that developmental bias, plasticity, niche construction and inheritance beyond the gene

deserve equal standing, against a position that they are already accommodated (Laland et al., 2015). This account takes the first side, and takes it as a position rather than as settled.

The gap is old and named. Natural selection explains the survival of the fittest and not the *arrival* of the fittest, which is de Vries's formulation and has stayed in use for over a century (de Vries, 1905; Wagner, 2014). It is not neglected — developmental constraint, evolvability and the biased structure of available variation are all active, and variation is known not to be isotropic: lineages move more readily along some directions than others, and the direction is a fact about the organism rather than about the pressure (Schluter, 1996). This account belongs on that side of the line and has nothing to add to the selection side.

The mechanism this account would offer there is named already, and the name is the same one: a lineage that has been changing does not stop when selection does. Maternal inheritance makes a population continue to evolve for many generations after selection ceases, sometimes in the direction selection opposes, and the effect is described as a form of *evolutionary momentum* in the follow-on treatment (Kirkpatrick and Lande, 1989; Lande and Kirkpatrick, 1990). A related effect has been measured in the field, where responses in red squirrels were influenced by the previous generation's selection as well as the current one — under selection that was ongoing rather than ceased (McAdam and Boutin, 2004). Rate itself is likewise treated as an inherited, autocorrelated quantity in molecular evolution (Thorne et al., 1998), and whether evolutionary patterns arise from variational properties of traits or lineages, or instead from ecology, is stated as the open question by the people working on it (Love et al., 2022).

So this is not a gap and nothing here fills one, and an earlier version of this passage claimed a remaining novelty that the momentum paper has since withdrawn: the published account's carrier is the maternal phenotype, which is present structure doing the work, so it holds the position this account holds rather than a rival to it. What the correspondence supplies is a grounding, not a contribution.

Need does not generate structure, and cannot: a need for legs is a relation to legs, and there are none yet to be related to. Need filters what has been produced, which is exactly selection's work. The producing is the part above, and if change were proportional to what is required nothing new would ever appear, since every change would be a closer fit to a demand already present. *Novelty requires overshoot* — structure has to exist before anything can find a use for it, so the capacity that looks wasteful is the only thing that has ever produced anything not already implied.

That inference is not new either, and one version of it holds in simulation. Search directed at an objective can be actively prevented from reaching it by that objective, and on two benchmark tasks — maze navigation and biped walking — search for novelty alone outperformed it (Lehman and Stanley, 2011). Those are evolved controllers rather than organisms, so this transfers a result across domains and does not supply biological evidence. Redundant capacity released from immediate requirement is the textbook biological route to novel function (Ohno, 1970), and complexity arising with no adaptive cause is an established position in molecular evolution (Stoltzfus, 1999; Lukeš et al., 2011). What is thin rather than absent is the tie to an overshooting *rate* specifically.

Three qualifications. Morphological stasis is not the absence of evolution and only the former is claimed. The mechanism of stasis *in general* is contested — stabilising selection (Estes and Arnold, 2007) and ephemeral divergence with gene flow (Futuyma, 2010), whose title names constraint as the other live account, are the developed positions — but there is no horseshoe-crab-specific three-way contest, and an earlier version of this remark implied one, and misdescribed the study that measured the group: it argues habitat and developmental constraint and *also* endorses stabilising selection explicitly, including in its conclusion. It fits no evolutionary-mode models at all, which is worth knowing beside Hunt et al. (2025) below. And settling is not safety: *Limulus* is presently threatened by harvesting and habitat loss, which is the environment moving faster than any trade was struck for.

A caution about stasis generally, since the remark leans on it. The punctuated reading is not the settled state of the evidence: a recent assessment finds stasis favoured in fewer than half of examined cases, with meandering random walks or more complex models accounting for most of the rest, which is why the citation above is worded as what the record *was read as* showing (Hunt et al., 2025).

Remark 21.6 (Cognition under the same treatment). Cognition is compressed by any measure — a brain takes on the order of a fifth of a body's resting energy for a fiftieth of its mass, and an energy budget for signalling has been computed in detail (Attwell and Laughlin, 2001) — for rodent grey matter, with the whole-body figures quoted there from earlier work rather than established by it. The stronger claim, that neural arrangements are forced by that budget rather than chosen, is Sterling and Laughlin (2015)'s thesis and not the budget paper's. Two constraints can then be named in the way the discipline above requires, and only two; neither is supplied by either citation, and both are ordinary neuroscience. Conduction takes time over distance, so there is no central synchroniser and coordination is local and coupled — the same concession distributed engineering made, for the same reason. And bandwidth is scarce, so what is not predicted is what gets sent.

Two further candidates are excluded. That cost is inherited along provision is this account's own thesis (Proposition 19.5), so offering it as an independent constraint on brains would be circular; and that introspection does not reach the machinery is not a constraint but the conclusion relabelled.

What this licenses is narrower than *cognition is the same*. Two structures under the same binding constraints land on the same arrangements over the region those constraints touch, and no further: cognition is not thereby computation, all cognitions are not thereby alike, and nothing here says a brain is a computer.

The consequence for this paper's own composition claim is the deflation stated in Section 1: if recurrence is what compressed structures do, what a recurrence evidences is the constraint and not the arrangement, and the composition falsifier is negative-only. The test that remains is not *a domain not under these constraints*, which is empty — every physical domain is under finite signal speed, finite bandwidth and a thermodynamic floor. It is a domain under *weak* compression: somewhere with slack, where cost pressure is not binding, and where the arrangement should therefore be absent. Ritual, ornament, and systems maintained past their purpose are candidates. Finding the arrangement there, rendered with equal facility, would show the vocabulary doing the work.

That proposition is the paper's discriminating test between privileged-carving realism and cost-shaped pluralism, and the systematics literature is where it can be run. The competing species concepts do converge, on a common notion of a separately evolving lineage, with the long dispute reinterpreted as disagreement over operational criteria rather than over the concept (de Queiroz, 2007) — and that convergence is what this account predicts rather than what embarrasses it. Whether it is *enforced* is the open question, and de Queiroz himself holds the boundaries to be real but epistemically hard, which is the opposing horn honestly stated; the criteria continue to disagree hardest over asexual organisms, bacteria exchanging genes horizontally and recent divergences. This paper does not claim the case is settled in its favour. It claims the case is a test.

The wider prediction is that classification degrades in a *located* way rather than uniformly, and it must be stated with the operationalisation that makes it a test.

Hypothesis 21.7 (Located degradation)**.** *Where a classification's purpose can be specified independently and in advance of inspecting its failures, the failures concentrate near the boundary of that purpose more than a uniform distribution over the domain would produce. The claim is comparative and is not that failures occur nowhere else.*

The advance-specification clause is what keeps this from circularity, and it is a real constraint:

a purpose read off from where the failures already are confirms nothing. Independent evidence exists elsewhere — classification systems in institutional use are documented as leaking wherever they have been examined closely, carrying residual categories and encoding their builders' purposes (Bowker and Star, 1999) — and that literature's concept of *torque* — the twisting of a person's own trajectory against a standardised classification that does not fit it — is a closer instrument for this hypothesis than leakage is, with the qualification that its subject is a biography rather than a use.

Two further results bear on the general claim without establishing it. No clustering function satisfies scale-invariance, richness and consistency together (Kleinberg, 2002) — but the same paper shows any two of the three are jointly achievable, so what is established is that the three cannot be had at once rather than that no cut is canonical. Kleinberg reads it that way himself, and dismisses a canonicity result for single-linkage cited on his second page as resting on a property that is an implicit optimisation criterion which single-linkage uniquely optimises; that dismissal is his, and is noted here rather than relied on. It is a trade-off theorem, not an impossibility of carving, and it is a deliberate transplant of Arrow's method rather than an independent convergence — the paper says so. And the question of which predicates are projectible has no syntactic answer (Goodman, 1955), which is usually reported as a defeat. Goodman's own criterion is *entrenchment*, a record of a predicate's having been *projected* before — a history of use and deliberately not a history of success, so it does not license reading entrenchment as tracking what has been cheap to run. The resemblance to a cost criterion is worth noticing and is not evidence for one.

The constructive response is this paper's own and needs no companion: if the trouble is a cut fixed in advance, do not fix one. Corollary 13.4 already gives the mechanism — recognition and growth as one operation, joined where the residual meets the grammar that admits it — so a category is minted only where what arrived will not fit, and the codebook grows rather than the cut being defended. That recovers no privileged carving, because there is none, and does not need to.

The position is not new, and its nearest neighbour differs in grounding rather than in conclusion. A pluralism admitting many equally legitimate and equally real classifications, none of them the one nature intended, is argued at length (Dupré, 1993); that argument rests on objective similarity structure, where this one rests on what a cut costs to run, which is the more deflationary footing and reintroduces a ranking over cuts that "equally legitimate" was formulated to deny. The convergence is on the negative thesis; the divergence in grounding is real and is not smoothed over here.

22 Determinism and apparent stochasticity

Proposition 22.1 (Graph-law determinism)**.** *For a fully specified antecedent configuration—including the graph state, operative transformation structure, admissibility grammar, and every causally operative containing relation relevant to the collapse—the realised continuation is determined:*

$$\boxed{\left(G_t, \mathcal{R}_t^{\text{causal}}\right) \mapsto G_{t+1}.}$$

The Graph Law contains no primitive random actualisation operator.

Proposition 22.1 is to be read through Proposition 17.5: the antecedent configuration is causally sufficient *present* structure, not a retained temporal history. Determinism is a relation between present structure and realised continuation, and Proposition 17.4 keeps it from being overstated — a determinable continuation is not thereby a determined one.

Apparent stochasticity arises when a frame does not represent all of the causally operative dimensions or relations required to determine the realised continuation. A probability distribution

may then be an appropriate model of that frame's uncertainty, but the probability belongs to the incomplete representation rather than to the substrate transition itself. Treating probability as a representation of incomplete information has a long statistical-mechanical and Bayesian lineage (Jaynes, 1957).

Thus

stochastic observation = deterministic dynamics viewed through unresolved causal structure.

Remark 22.2 (Direction of the borrowing)*.* The order of discovery ran from quantum measurement outward, not from a general primitive inward. Collapse was adopted because compatibility with quantum measurement is a constraint on any theory of realised change, and was then assumed to hold generally. The bridge is therefore stated conditionally because the *generalisation* is the conjecture, not because the compatibility is in doubt. This is the structure of a promoted constraint — a fact that must hold in one regime, assumed to hold everywhere, with the consequences derived and exposed to falsification — and it is why Section 24 carries a falsifier for the bridge specifically. Origin does not establish a claim; the falsifier is what does the work either way.

The distinction of Remark 17.2 is what makes quantum measurement the visible case for the conditional collapse bridge (Remark 22.2). In that regime the pre-measurement alternatives provably cannot be re-read as an *undisclosed* already-realised classical value: Bell's theorem excludes the local hidden-variable class and Kochen–Specker constrains noncontextual value assignment (Bell, 1964; Kochen and Specker, 1967). What remains is *undetermined* in the strict sense — not a value withheld, but no value yet. Elsewhere the two readings are usually interchangeable in practice; there they are not.

Deterministic quantum models demonstrate that empirical quantum statistics do not by themselves force fundamental stochasticity: Bohm's hidden-variable construction is explicitly deterministic (Bohm, 1952), while the constraints just cited rule out the local and noncontextual hidden-variable classes rather than determinism as such. Graphic Equalisation does not identify itself with Bohmian mechanics; these results establish only that determinism must be distinguished from locality and noncontextuality.

A stochastic law may be bolted onto a grounding as an epistemic predictive model over unresolved continuations. Doing so is an additional modelling layer:

$$\text{deterministic Graph Law} \longrightarrow \text{frame-limited uncertainty model},$$

not a modification of the Graph Law into a stochastic substrate.

23 Positioning and prior art

The individual ingredients of Graphic Equalisation have substantial prior art. The proposed contribution is their conjunction at one minimal substrate level and the additional distinctions forced by making that conjunction executable.

Graph rewriting and typed transformation. Algebraic graph transformation provides mature formalisms for typed graphs, rule application, partiality, and structural consistency (Ehrig et al., 2006). Graphic Equalisation adopts graph transformation as substrate vocabulary but makes frame-relative boundary forcing, collapse, and identity part of the same dynamical account.

Reflection and self-modification. Computational reflection establishes that a system can represent and modify meta-level structure governing its own execution (Smith, 1982; Maes, 1987). The specific identity "the grammar acting on itself is the loop" is the present paper's claim, not a result attributed to those sources.

Cybernetics and feedback. Classical cybernetics and control theory distinguish open-loop dynamics from systems in which error is returned into subsequent control (Wiener, 1948; Ashby, 1956). The residual construction generalises the same causal distinction: mismatch changes later action only where it survives as represented state.

Quantum measurement and relational description. The separation of pre-realisation state from realised measurement record is standard in the measurement problem (von Neumann, 1955). Everett's relative-state formulation and Rovelli's relational quantum mechanics provide important precedents for refusing a single privileged observer-independent account of all facts (Everett, 1957; Rovelli, 1996). Graphic Equalisation uses these as neighbouring structures rather than deriving itself from any one interpretation.

Deterministic quantum theories. Bohm's 1952 model is a direct proof of principle that deterministic dynamics can reproduce nonrelativistic quantum predictions when supplemented by additional state (Bohm, 1952). Bell and Kochen–Specker impose decisive constraints on local and noncontextual hidden-variable theories (Section 22). Consequently, the determinism claim in Proposition 22.1 should be read as a commitment requiring an adequate nonlocal/contextual grounding where quantum phenomena demand it, not as a revival of unconstrained classical hidden variables.

Causal order. Causal-set theory supplies a well-developed example in which causal order is taken as primitive while continuum metric structure is additional or emergent (Bombelli et al., 1987). Graphic Equalisation makes only the weaker substrate claim that ordinal causal order does not itself entail a real-valued metric time.

Emergent and relational time. That time is not a substrate dimension (Section 18) has substantial physics precedent, and the present account sits at the most-local end of it. The Wheeler–DeWitt equation of canonical quantum gravity carries no time parameter (DeWitt, 1967) — the "problem of time" this substrate dissolves rather than solves, there having been no time variable to recover. The Page–Wootters mechanism derives apparent evolution from entanglement between a clock subsystem and the rest (Page and Wootters, 1983), closely matching "time inferred from the relations a frame holds," but retains a single global state where this account is per-frame. Barbour's Machian relationalism takes time to be abstracted from change rather than a container (Barbour, 1999), and Rovelli's relational and thermal-time programme refuses a privileged global time (Rovelli, 1996, 2011). The distinguishing move here is to keep none of the global objects these programmes retain — global wavefunction, global configuration space, global state — reading time entirely from local change.

Computational-universe and graph-rewrite models. The closest extant framework is the Wolfram Physics Project, in which the universe is a hypergraph evolved by rewrite rules, time is progressive rule application, and observer-independent structure follows from *causal invariance* — different update orderings yielding the same causal graph (Wolfram, 2002, 2020). The structural agreement is deep: graph plus rewrite rules, time emergent from rule application rather than posited, order-independence, and observer-relative slicing. Four differences locate the present contribution. First, Γ is an admissibility grammar that is itself graph data and self-modifying (the grammar-is-the-loop theorem), whereas Wolfram fixes a rule per universe or takes the entire rule space — the Ruliad — at once. Second, collapse realises *one* frame-relative continuation, whereas Wolfram retains all branches (the multiway graph) and derives quantum structure from them. Third, residual and growth-by-minting from it have no counterpart in a fixed rule. Fourth, the present substrate keeps no global object — no single hypergraph being updated — being local and frame-relative throughout, and its claims are information-level and

cross-domain rather than an attempt to derive our specific universe. This last is why c falls out here as a *per-frame* unit (developed in the companion physics-grounding paper): sub-rules with closure give each frame its own cycle and hence its own $c = 1$, invariant by construction, whereas a single global update yields c only as an emergent speed limit shown observer-independent by causal invariance — the constant, but not the per-frame machinery beneath it. That an independent programme reached the shared shape from cellular automata and physics, while this one reached it from cognition and engineering, is evidence for the arrangement claim of Section 1 rather than against its novelty.

Information and minimum description. The preference for the smallest sufficient ontology follows the same methodological direction as information-theoretic and minimum-description-length approaches: do not introduce representational structure that buys no additional explanatory or predictive distinction (Shannon, 1948; Rissanen, 1978). This motivates but does not prove the Discrete Sufficiency Hypothesis.

Standing beside, not replacing. Graphic Equalisation is not proposed as a replacement for established theories. It is intended to sit beneath or beside them as a substrate description. A grounding may add continuous geometry, conservation laws, quantum amplitudes, relativistic metrics, biological mechanisms, institutional rules, or probabilistic models of unresolved lower-frame observations. In particular, relational quantum mechanics (Rovelli, 1996), Everettian relative-state quantum mechanics (Everett, 1957), and deterministic Bohmian mechanics (Bohm, 1952) make mutually different commitments while remaining compatible with large parts of standard quantum phenomenology; this paper does not require choosing among those complete physical models at the substrate-definition stage. Such additions do not alter Proposition 22.1: within the Graph Law, realised transition remains deterministic once the causally complete antecedent configuration is specified.

The substrate claim remains

represented state → admissible possibility → collapse → realised state.

24 Falsifiability commitments

Each commitment below is marked [SHAPE] or [DETAIL]. A [DETAIL] failing is a revision: the account survives with that mechanism replaced. A [SHAPE] failing is a refutation, because what fails is something the substrate is claimed to require or to permit. The distinction is the one Section 1 rests its epistemic claim on, and marking it here is what makes that claim checkable rather than asserted.

[shape] Composition. Prior to the component commitments below: the arrangement claim of Section 1 fails if the structure's recurrence across unrelated domains is an artefact of description rather than a property of those domains. Evidence against it would be a demonstration that any sufficiently rich domain can be rendered in this vocabulary with equal facility, including domains chosen adversarially to lack the structure. Evidence for it is independent re-derivation, in a domain's own idiom, without reference to this theory. The burden here is deliberately on the negative side, because that side is closeable: a single adversarial domain rendered with equal facility settles it against the claim. No requirement is placed on the positive side to exhibit domains that share *nothing* — every pair of domains shares something, so that standard is unmeetable in principle, and discharging it would mean enumerating domains until an assessor chose to stop. A test whose stopping condition lies with the assessor rather than with the claim is not a test, and this paper does not offer one.

1. [DETAIL] **Solutions are nearly unique under a binding constraint.** The posit under Propositions 21.1 and 21.2. Fails if structures optimising hard against the same binding constraint, over the same region, are exhibited arriving at arrangements that do not resemble each other — in which case constraint narrows the space without selecting within it, and both propositions weaken to that.

2. [DETAIL] **Located degradation.** Fails if, for classifications whose purpose is specifiable in advance of inspecting their failures, failure cases are distributed no differently near the purpose boundary than across the rest of the domain (Hypothesis 21.7).

3. [DETAIL] **Convergence is not enforced.** Fails if convergence between independently developed classifications of one domain extends to the cases where the classifications' purposes differ, which is what a privileged boundary would produce and cost-shaped convergence would not (Proposition 21.1).

4. [SHAPE] **Minimal graph sufficiency.** Fails if basic realised graph dynamics require a primitive beyond (g, r, Γ) rather than represented or relational structure.

5. [SHAPE] **Discrete sufficiency.** Fails if a claimed observable consequence cannot be preserved by any discrete representation, including dynamically varying quantisation, but is preserved by a genuinely non-discrete primitive.

6. [SHAPE] **Antecedent-state principle.** Fails if a realised transition can require as antecedent a fact that exists only as consequence of that same transition, absent prior fixed-point structure.

7. [SHAPE] **Graph-law determinism.** Fails if two genuinely identical, causally complete antecedent configurations produce different realised successors without any additional causally operative distinction. Variation observed only from a frame that omits determining dimensions does not constitute such a falsification.

8. [SHAPE] **Admissibility is not armour.** Fails if self-admissibility must constrain arbitrary containing-world forcing.

9. [SHAPE] **No implicit historical state.** Fails if an unrepresented past event alters later transition structure despite leaving no causal trace in graph or boundary state.

10. [SHAPE] **Frame-relative identity.** Fails if sameness and persistence are derivable from (g, r, Γ) alone without individuation.

11. [SHAPE] **Higher-frame first-class structure.** Fails if a containing graph cannot possess ordinary relational state about a lower graph unless the same quantity is primitive inside the lower graph.

12. [SHAPE] **Residual is inadmissibility.** Fails if a graph-originated deformation can be left as residual by a containing frame that nonetheless admits it under Γ_{W_c}, or absorbed while inadmissible, so that residual and non-admission come apart.

13. [DETAIL] **Cost is the write.** Fails if a substrate operation other than writing a delta carries an irreducible cost — if reading, holding a standing structure, or the boundary relation itself must be paid for with a cost that is not the cost of the write they occasion.

14. [SHAPE] **Self-change is subset-change.** The claim is that a graph changes itself and its subgraphs directly (rule-governed), and affects its container only as a subset of it, by propagation through coupling. Fails if a graph changes its container by a means other than changing the region of it that the graph is — a container-change not propagating from the graph's own subset — or if a graph can self-terminate without acting on the container.

15. [SHAPE] **Minting is a rule.** Fails if grammar-extension (growth) requires a primitive operation beneath the rules rather than a reflective rule subject to the same economics, admissibility, and antecedent-state principle — in particular, if a graph can mint its first distinctions without any seeded from its container.

16. [SHAPE] **Partial-order sufficiency.** Fails if realised dynamics require a total order over collapses in distinct frames — a global simultaneity — rather than only the boundary-induced partial order.

17. [SHAPE] **Globally-async/locally-sync.** Fails if coupling never induces local synchrony (boundary-related collapses remain unordered) or if synchrony is forced without coupling (a global instant exists independent of any boundary).

18. [DETAIL] **Provenance is never closed by inference.** Not that primitives fail to leak — they do leak, and Proposition 19.2 says so. This fails if a frame can *settle* the provenance of its own operator set from inside: if some in-frame procedure establishes, rather than merely evidences, whether its operators are self-supplied, how many distinct structures supply them, or whether a serial hidden supply underlies an apparently asynchronous frame The escape is not that further primitives were used, which is unavoidable, but that their provenance is open in the same way, so no procedure closes the question rather than relocating it.

25 Conclusion

The theory is deliberately small. A graph need not know what it is, where it is, what will happen, or what it wants. Present structure determines admissible continuation; interaction and collapse produce realised history; realised history changes the antecedent from which subsequent possibility is generated.

Recursion requires no external homunculus. Where the grammar acts on its own represented structure, the grammar is the loop. World forcing is not self-action, residual is discrepancy rather than diagnosis, history matters only where it survives as present structure, and identity exists only under individuation by a frame.

Pressed for what is compulsory, the theory keeps almost nothing. What survives every attempted removal is structure with relations among its parts; what rules running adds is that anything *happens* (Remark 3.2), which is the condition of everything this account is about and is not the condition of there being anything; the specific laws above it — inertia, closure, the shape of collapse, and the write-economics itself — are contingent rule content rather than necessities, and not strict universals (Remark 12.10), which is the sense in which this substrate underlies groundings without dictating them.

Every mechanism used here belongs to someone else and is cited as such; the claim is the arrangement (Section 1), and the method by which the arrangement was reached is removal rather than addition — of primitive time, of a privileged frame, of a cost that is the same whatever runs the rules. Removals are small to state and large in consequence, because results that had been independently working around a deleted assumption stop being independent.

The account forbids one thing, and Section 20 states it and its formal analogue: a frame cannot get outside itself. Everything else that has looked impossible has so far turned out to be contingent rule content that some grounding happens not to instantiate — which is a measure of how much had been filed as necessary without being checked, and is not itself evidence for the account, since the enumeration is open and this paper does not offer an open enumeration as a test (Section 24).

The regress in that prohibition has both ends, and neither is safe to state alone. A superframe dissolves the frame below's limit: holding both of two locations it has units closing across them, and quantities the subframes could not obtain are ordinary in its terms. And a superframe is

a frame, so it reproduces the same blindness about its own running. Ascent buys something real and never buys everything — the limit is dissolved locally at every step and terminated at none, which is why no amount of it yields a view from nowhere. What ascent yields is a different somewhere.

References

W. Ross Ashby. *An Introduction to Cybernetics*. Chapman & Hall, London, 1956.

David Attwell and Simon B. Laughlin. An energy budget for signaling in the grey matter of the brain. *Journal of Cerebral Blood Flow and Metabolism*, 21(10):1133–1145, 2001.

John C. Avise, William S. Nelson, and Hiroaki Sugita. A speciational history of "living fossils": molecular evolutionary patterns in horseshoe crabs. *Evolution*, 48(6):1986–2001, 1994.

Julian Barbour. *The End of Time: The Next Revolution in Physics*. Oxford University Press, Oxford, 1999.

John S. Bell. On the Einstein Podolsky Rosen paradox. *Physics Physique Fizika*, 1:195–200, 1964.

Charles H. Bennett. Logical reversibility of computation. *IBM Journal of Research and Development*, 17(6):525–532, 1973.

Elwyn R. Berlekamp, John H. Conway, and Richard K. Guy. *Winning Ways for Your Mathematical Plays*. Academic Press, London, 1982. (Universality of Conway's Game of Life.)

Russell D. C. Bicknell, Julien Kimmig, Graham E. Budd, David A. Legg, Cheyenne Bader, Joachim T. Haug, André Kaiser, Petr Laibl, Yara Tashman, and Nicolás E. Campione. Habitat and developmental constraints drove 330 million years of horseshoe crab evolution. *Biological Journal of the Linnean Society*, 136(1):155–172, 2022.

David Bohm. A suggested interpretation of the quantum theory in terms of "hidden" variables. I and II. *Physical Review*, 85:166–193, 1952.

Luca Bombelli, Joohan Lee, David Meyer, and Rafael D. Sorkin. Space-time as a causal set. *Physical Review Letters*, 59:521–524, 1987.

R. Randal Bollinger, Andrew S. Barbas, Errol L. Bush, Shu S. Lin, and William Parker. Biofilms in the large bowel suggest an apparent function of the human vermiform appendix. *Journal of Theoretical Biology*, 249(4):826–831, 2007.

Geoffrey C. Bowker and Susan Leigh Star. *Sorting Things Out: Classification and Its Consequences*. MIT Press, Cambridge, MA, 1999.

Eric A. Brewer. Towards robust distributed systems. In *Proceedings of the 19th ACM Symposium on Principles of Distributed Computing (PODC)*, keynote, 2000.

K. R. Castellano and colleagues. Genome assembly of a living fossil, the Atlantic horseshoe crab *Limulus polyphemus*, reveals lineage-specific whole-genome duplications, transposable element-based centromeres, and a ZW sex chromosome system. *Molecular Biology and Evolution*, 42(2):msaf021, 2025.

Daniel M. Chapiro. *Globally-Asynchronous Locally-Synchronous Systems*. PhD thesis, Stanford University, 1984.

Jennifer A. Clack. *Gaining Ground: The Origin and Evolution of Tetrapods*. Indiana University Press, Bloomington, 2nd edition, 2012.

Richard Cloutier, Alice M. Clement, Michael S. Y. Lee, Roxanne Béchard, Vincent Roy, John A. Long, and Isabelle Bégin. *Elpistostege* and the origin of the vertebrate hand. *Nature*, 579:549–554, 2020.

Michael I. Coates and Jennifer A. Clack. Polydactyly in the earliest known tetrapod limbs. *Nature*, 347:66–69, 1990.

Michael I. Coates and Jennifer A. Clack. Fish-like gills and breathing in the earliest known tetrapod. *Nature*, 352:234–236, 1991.

Kevin de Queiroz. Species concepts and species delimitation. *Systematic Biology*, 56(6):879–886, 2007.

Hugo de Vries. *Species and Varieties: Their Origin by Mutation*. Open Court, Chicago, 2nd edition, 1905.

Bryce S. DeWitt. Quantum theory of gravity. I. The canonical theory. *Physical Review*, 160:1113–1148, 1967.

Blake V. Dickson, Jennifer A. Clack, Timothy R. Smithson, and Stephanie E. Pierce. Functional adaptive landscapes predict terrestrial capacity at the origin of limbs. *Nature*, 589:242–245, 2021.

James DiFrisco and Richard Gawne. Biological agency: a concept without a research program. *Journal of Evolutionary Biology*, 38(2):143–156, 2025.

John Dupré. *The Disorder of Things: Metaphysical Foundations of the Disunity of Science*. Harvard University Press, Cambridge, MA, 1993.

Hartmut Ehrig, Karsten Ehrig, Ulrike Prange, and Gabriele Taentzer. *Fundamentals of Algebraic Graph Transformation*. Springer, Berlin, 2006.

Niles Eldredge and Stephen Jay Gould. Punctuated equilibria: an alternative to phyletic gradualism. In *Models in Paleobiology*, pages 82–115. Freeman Cooper, San Francisco, 1972.

Suzanne Estes and Stevan J. Arnold. Resolving the paradox of stasis. *The American Naturalist*, 169(2):227–244, 2007.

Hugh Everett III. "Relative state" formulation of quantum mechanics. *Reviews of Modern Physics*, 29:454–462, 1957.

Douglas J. Futuyma. Evolutionary constraint and ecological consequences. *Evolution*, 64(7):1865–1884, 2010.

Kurt Gödel. Über formal unentscheidbare Sätze der Principia Mathematica und verwandter Systeme I. *Monatshefte für Mathematik und Physik*, 38:173–198, 1931.

Nelson Goodman. *Fact, Fiction, and Forecast*. Harvard University Press, Cambridge, MA, 1955.

Stephen Jay Gould and Elisabeth S. Vrba. Exaptation — a missing term in the science of form. *Paleobiology*, 8(1):4–15, 1982.

Gene Hunt, Kjetil L. Voje, and Lee Hsiang Liow. Punctuated equilibrium: state of the evidence. *Paleobiology*, 51(4):652–661, 2025.

E. T. Jaynes. Information theory and statistical mechanics. *Physical Review*, 106:620–630, 1957.

K. Ingemar Jönsson, Elke Rabbow, Ralph O. Schill, Mats Harms-Ringdahl, and Petra Rettberg. Tardigrades survive exposure to space in low Earth orbit. *Current Biology*, 18(17):R729–R731, 2008.

Mark Kirkpatrick and Russell Lande. The evolution of maternal characters. *Evolution*, 43(3):485–503, 1989.

Jon Kleinberg. An impossibility theorem for clustering. In *Advances in Neural Information Processing Systems 15*, pages 463–470, 2002.

Simon Kochen and Ernst P. Specker. The problem of hidden variables in quantum mechanics. *Journal of Mathematics and Mechanics*, 17:59–87, 1967.

Paul Kocher, Jann Horn, Anders Fogh, Daniel Genkin, Daniel Gruss, Werner Haas, Mike Hamburg, Moritz Lipp, Stefan Mangard, Thomas Prescher, Michael Schwarz, and Yuval Yarom. Spectre attacks: Exploiting speculative execution. In *2019 IEEE Symposium on Security and Privacy*, pages 1–19, 2019.

Saul Kripke. Outline of a theory of truth. *Journal of Philosophy*, 72(19):690–716, 1975.

Kevin N. Laland, Tobias Uller, Marcus W. Feldman, Kim Sterelny, Gerd B. Müller, Armin Moczek, Eva Jablonka, and John Odling-Smee. The extended evolutionary synthesis: its structure, assumptions and predictions. *Proceedings of the Royal Society B*, 282(1813):20151019, 2015.

Leslie Lamport. Time, clocks, and the ordering of events in a distributed system. *Communications of the ACM*, 21(7):558–565, 1978.

James C. Lamsdell. Horseshoe crab phylogeny and independent colonizations of fresh water: ecological invasion as a driver for morphological innovation. *Palaeontology*, 59(2):181–194, 2016.

Rolf Landauer. Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development*, 5(3):183–191, 1961.

Russell Lande and Mark Kirkpatrick. Selection response in traits with maternal inheritance. *Genetical Research*, 55(3):189–197, 1990.

F. William Lawvere. Diagonal arguments and cartesian closed categories. In *Category Theory, Homology Theory and their Applications II*, Lecture Notes in Mathematics 92, pages 134–145. Springer, 1969.

Joel Lehman and Kenneth O. Stanley. Abandoning objectives: evolution through the search for novelty alone. *Evolutionary Computation*, 19(2):189–223, 2011.

Jonathan B. Losos. *Improbable Destinies: Fate, Chance, and the Future of Evolution*. Riverhead, New York, 2017.

Alan C. Love, Mark Grabowski, David Houle, Lee Hsiang Liow, Arthur Porto, Masahito Tsuboi, Kjetil L. Voje, and Gene Hunt. Evolvability in the fossil record. *Paleobiology*, 48(2):186–209, 2022.

Julius Lukeš, John M. Archibald, Patrick J. Keeling, W. Ford Doolittle, and Michael W. Gray. How a neutral evolutionary ratchet can build cellular complexity. *IUBMB Life*, 63(7):528–537, 2011.

Pattie Maes. Concepts and experiments in computational reflection. In *Proceedings of OOPSLA '87*, 1987.

Andrew G. McAdam and Stan Boutin. Maternal effects and the response to selection in red squirrels. *Proceedings of the Royal Society B*, 271(1534):75–79, 2004.

George R. McGhee. *Convergent Evolution: Limited Forms Most Beautiful*. MIT Press, Cambridge, MA, 2011.

Wenyan Nong and colleagues. Horseshoe crab genomes reveal the evolution of genes and microRNAs after three rounds of whole genome duplication. *Communications Biology*, 4:83, 2021.

F. John Odling-Smee, Kevin N. Laland, and Marcus W. Feldman. *Niche Construction: The Neglected Process in Evolution*. Princeton University Press, Princeton, NJ, 2003.

Susumu Ohno. *Evolution by Gene Duplication*. Springer, Berlin, 1970.

Dag Arne Osvik, Adi Shamir, and Eran Tromer. Cache attacks and countermeasures: The case of AES. In *Topics in Cryptology — CT-RSA 2006*, pages 1–20. Springer, 2006.

Don N. Page and William K. Wootters. Evolution without evolution: Dynamics described by stationary observables. *Physical Review D*, 27:2885–2892, 1983.

Jorma Rissanen. Modeling by shortest data description. *Automatica*, 14:465–471, 1978.

Tomoko Ohta. Slightly deleterious mutant substitutions in evolution. *Nature*, 246(5428):96–98, 1973.

Michael Lynch and John S. Conery. The origins of genome complexity. *Science*, 302(5649):1401–1404, 2003.

Kenneth D. Whitney and Theodore Garland Jr. Did genetic drift drive increases in genome complexity? *PLoS Genetics*, 6(8):e1001080, 2010.

Carlo Rovelli. Relational quantum mechanics. *International Journal of Theoretical Physics*, 35:1637–1678, 1996.

Carlo Rovelli. Forget time. *Foundations of Physics*, 41:1475–1490, 2011.

Sophie Sanchez, Paul Tafforeau, Jennifer A. Clack, and Per E. Ahlberg. Life history of the stem tetrapod *Acanthostega* revealed by synchrotron microtomography. *Nature*, 537:408–411, 2016.

Dolph Schluter. Adaptive radiation along genetic lines of least resistance. *Evolution*, 50(5):1766–1774, 1996.

Claude E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27:379–423, 623–656, 1948.

Brian Cantwell Smith. *Reflection and Semantics in a Procedural Language*. MIT Laboratory for Computer Science, 1982.

Heather F. Smith, William Parker, Sanet H. Kotzé, and Michel Laurin. Multiple independent appearances of the cecal appendix in mammalian evolution and an investigation of related ecological and anatomical factors. *Comptes Rendus Palevol*, 12(6):339–354, 2013.

Peter Sterling and Simon Laughlin. *Principles of Neural Design*. MIT Press, Cambridge, MA, 2015.

Thomas A. Stewart, Justin B. Lemberg, Ailis Daly, Edward B. Daeschler, and Neil H. Shubin. A new elpistostegalian from the Late Devonian of the Canadian Arctic. *Nature*, 608:563–568, 2022.

Arlin Stoltzfus. On the possibility of constructive neutral evolution. *Journal of Molecular Evolution*, 49(2):169–181, 1999.

Alfred Tarski. Der Wahrheitsbegriff in den formalisierten Sprachen. *Studia Philosophica*, 1:261–405, 1936.

Jeffrey L. Thorne, Hirohisa Kishino, and Ian S. Painter. Estimating the rate of evolution of the rate of molecular evolution. *Molecular Biology and Evolution*, 15(12):1647–1657, 1998.

Rose Trappes, Behzad Nematipour, Marie I. Kaiser, Ulrich Krohs, Koen J. van Benthem, Ulrich R. Ernst, Jürgen Gadau, Peter Korsten, Joachim Kurtz, Holger Schielzeth, Tim Schürmann, and Claudia Zeiss. How individualized niches arise: defining mechanisms of niche construction, niche choice, and niche conformance. *BioScience*, 72(6):538–548, 2022.

Alan M. Turing. On computable numbers, with an application to the Entscheidungsproblem. *Proceedings of the London Mathematical Society*, s2-42:230–265, 1936.

John von Neumann. *Mathematical Foundations of Quantum Mechanics*. Princeton University Press, Princeton, 1955. English translation of the 1932 German edition.

Andreas Wagner. *Arrival of the Fittest: Solving Evolution's Greatest Puzzle*. Current, New York, 2014.

Norbert Wiener. *Cybernetics: Or Control and Communication in the Animal and the Machine*. MIT Press, Cambridge, MA, 1948.

Stephen Wolfram. *A New Kind of Science*. Wolfram Media, Champaign, IL, 2002.

Stephen Wolfram. *A Project to Find the Fundamental Theory of Physics*. Wolfram Media, Champaign, IL, 2020.

Noson S. Yanofsky. A universal approach to self-referential paradoxes, incompleteness and fixed points. *Bulletin of Symbolic Logic*, 9(3):362–386, 2003.